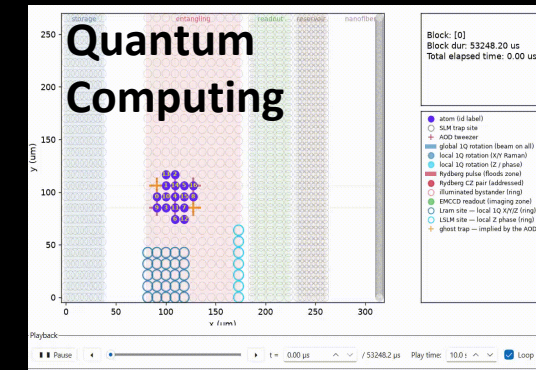
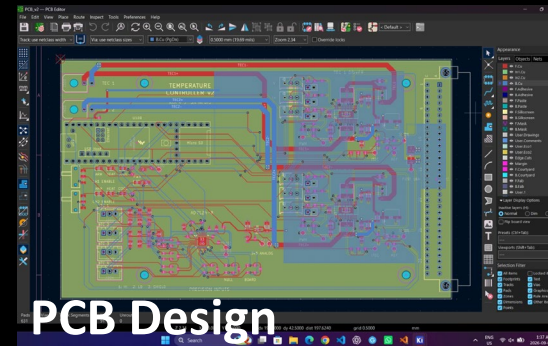
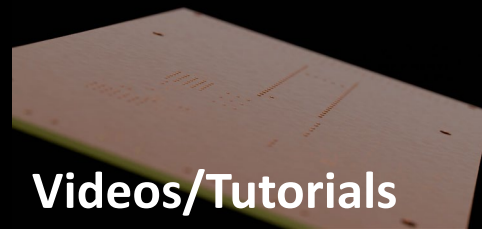


Modern Experimental Techniques

2) Electronics for experimentalists



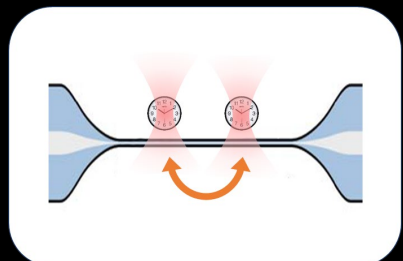
Shayne Bennetts

s.p.bennetts@g.iams.sinica.edu.tw

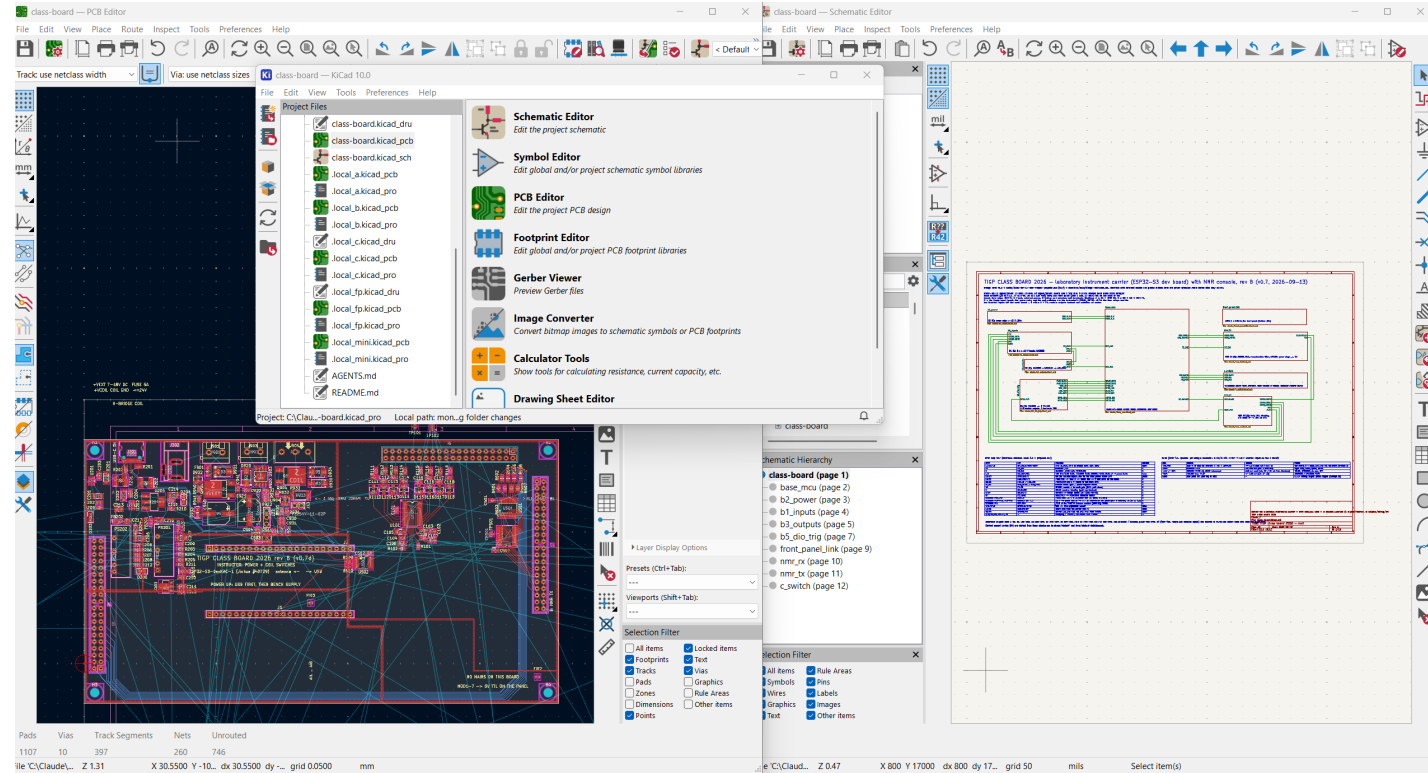
IAAMS Yb Lab

Institute of Atomic and Molecular Sciences
Academia Sinica, Taiwan

<https://iamsquantum.github.io>



- Laptop with 10 GB free, VS Code, Claude Code signed in
- KiCad 10 installed and opened once — Help → About shows 10.x
- hardware/class-board.kicad_pro opens, no missing-library warning
- A free JLCPCB account
- A USB-C data cable for the ESP32 — the instructor has spares
- WiFi details in the room



The "AI method" in 5 steps:

1. Vision
2. Specification/Plan
 - Ask the AI to ask you questions before it writes
3. Review (in a fresh Claude session)
 - Iterate — the first plan is never the last plan
4. Code
5. Test and verify

Never trust AI without testing and verifying

Projects 1 and 2 are yours - improve them, build more 😊

Project wall — Modern Experimental Techniques 2026

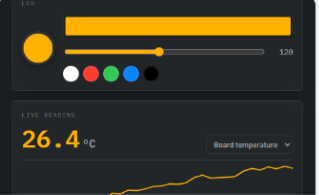
Every student's projects as they appear. Add yours: one JSON file and one picture in [the showcase repository](#) (the tutor does it with you). This page refreshes itself.

[All](#) [Project 1 - web app](#) [Project 2 - phone → ESP32](#) [Project 3 - the instrument](#) [Extras](#)



PROJECT 1 - WEB APP
Spin Runner + Pixel Plumber
Yi-Tsai Liang · @tojestoacja · 2026-09-12
Two canvas games. Spin Runner puts you inside a 500 MHz magnet as the magnetization itself: 64 isochromats stepped through the Bloch equations, so resonance, T1, T2* and the spin echo are simulated, not animated, and the run ends in the Fourier transform of the FID you recorded. The echo peaks at 35 ms, not the textbook $2\tau = 40$ ms, because the falling T2 envelope drags the maximum earlier.

[Open the app](#) [Project site](#) [Code](#)



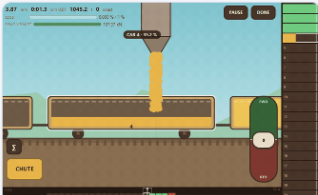
PROJECT 2 - PHONE → ESP32
Glowmeter
Tee Ren Qian · @Rengian03 · 2026-09-11
A phone web app that sets an ESP32-S3's onboard LED (colour + brightness) and streams a live measurement back as a strip chart at 20 Hz, over one WebSocket, no server in between. Runs unmodified class-board-2026 firmware; only the phone app is new, and it's flashed onto the board's own filesystem so it's served locally, avoiding any HTTPS/WebSocket mixed-content issue.

[Open the app](#) [Project site](#) [Code](#)



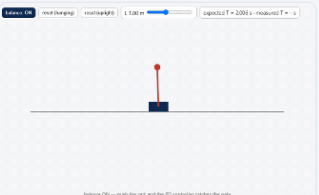
PROJECT 1 - WEB APP
Unfair Golf
Tee Ren Qian · @Rengian03 · 2026-09-11
An 8-hole browser golf game where every hole breaks a different rule of physics instead of just adding a hazard: gusting wind, a near-frictionless green, an inverse-square gravity well, teleporter pads, a drifting cup, and mirrored controls. Every wall bounce also carries randomized restitution, so no rebound is ever quite predictable.

[Open the app](#) [Project site](#) [Code](#)



PROJECT 1 - WEB APP
Load the train
Shayne Bennetts (instructor) · @shaynebennetts · 2026-09-10
Drive a locomotive and twenty hopper cars under a grain chute and fill them without spilling: grain leaves the chute with no forward speed, so every tonne you load drags the train back, harder the faster you go. The surprise is that the locomotive can outrun its own chute — full power settles at 6.2 m/s, but a car only reaches 90 % in one pass below 4.8 m/s.

[Open the app](#) [Project site](#) [Code](#)



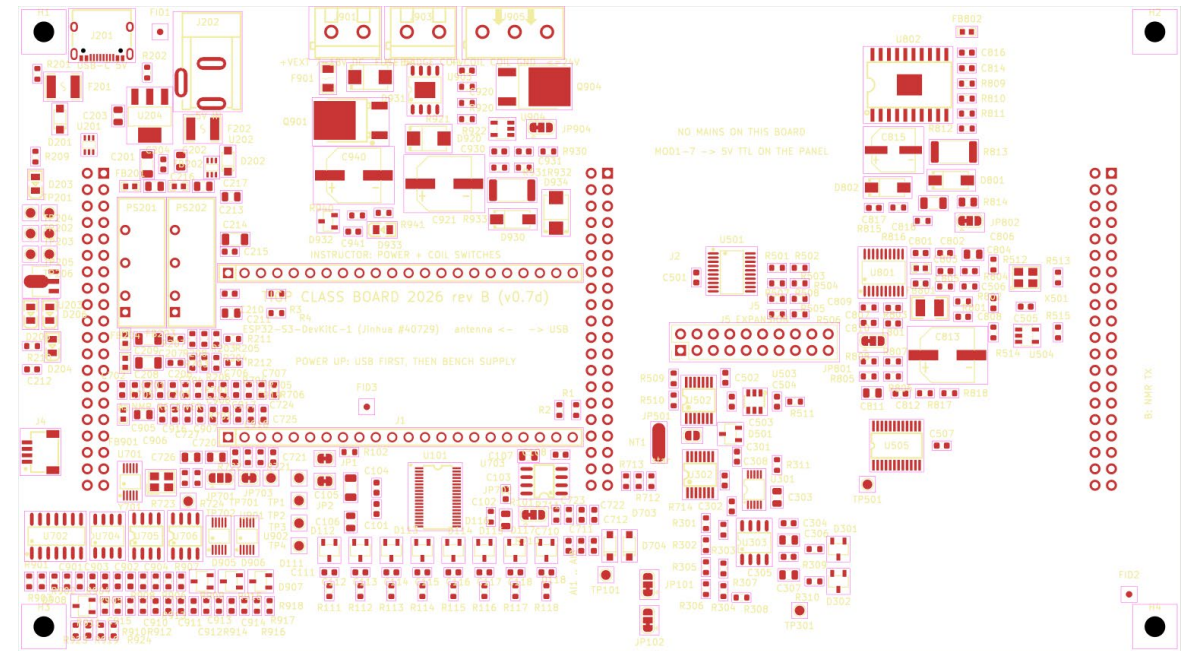
PROJECT 1 - WEB APP
Cart-pole — the reference example
Shayne Bennetts (instructor) · @shaynebennetts · 2026-09-06
The instructor's reference for a Project-1 build: a cart-pole you push with the arrow keys, or leave to a PD controller to balance. It exists to show a simulation being checked — predicted $2\pi\sqrt{L/g} = 2.006$ s against 1.987 s measured, with the cart recoil and the finite amplitude accounting for the difference to 0.1 %.

[Open the app](#) [Project site](#) [Code](#)

TiGP Workshop 2 outline



- Setup check — KiCad running
- Electronics 101 — enough to understand what we are building
- Choosing chips is the design
- PCB 101, and the six KiCad operations
- AI in the electronics pipeline
- Project 3a — build the class board together

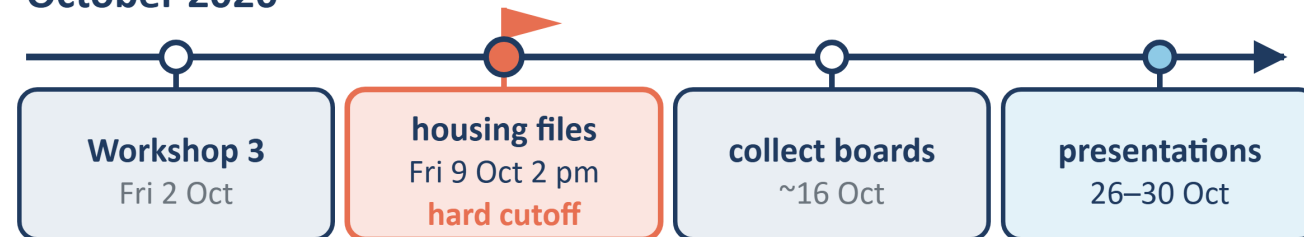


- Mon 28 Sep, 2 pm — the class PCB order goes to the factory
- Fri 9 Oct, 2 pm — housing files for printing and cutting
- ~16 Oct — collect the assembled boards and the housings
- 26–30 Oct — presentation and demonstration, ~15 minutes each
- Fri 25 Sep is a public holiday; Workshop 3 is Fri 2 Oct

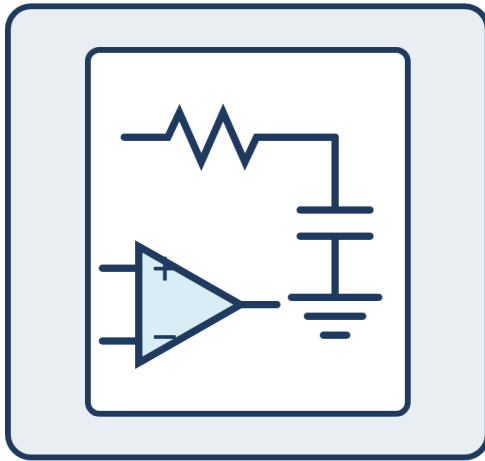
September 2026



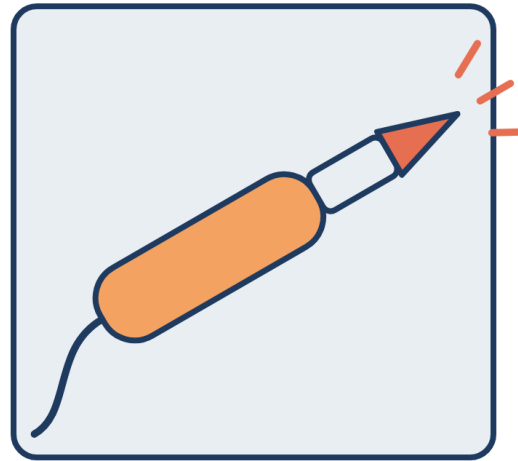
October 2026



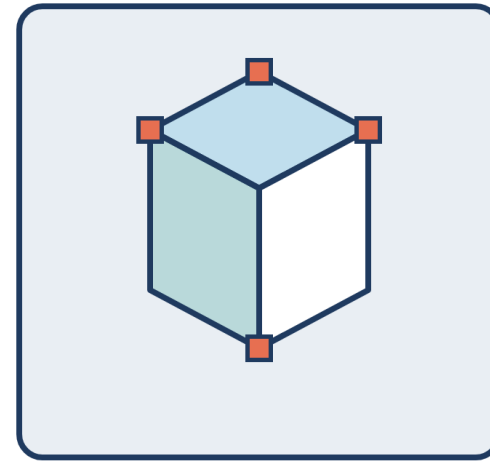
- Have you read a schematic before?
- Have you soldered?
- Have you used a CAD program?



Read a schematic?

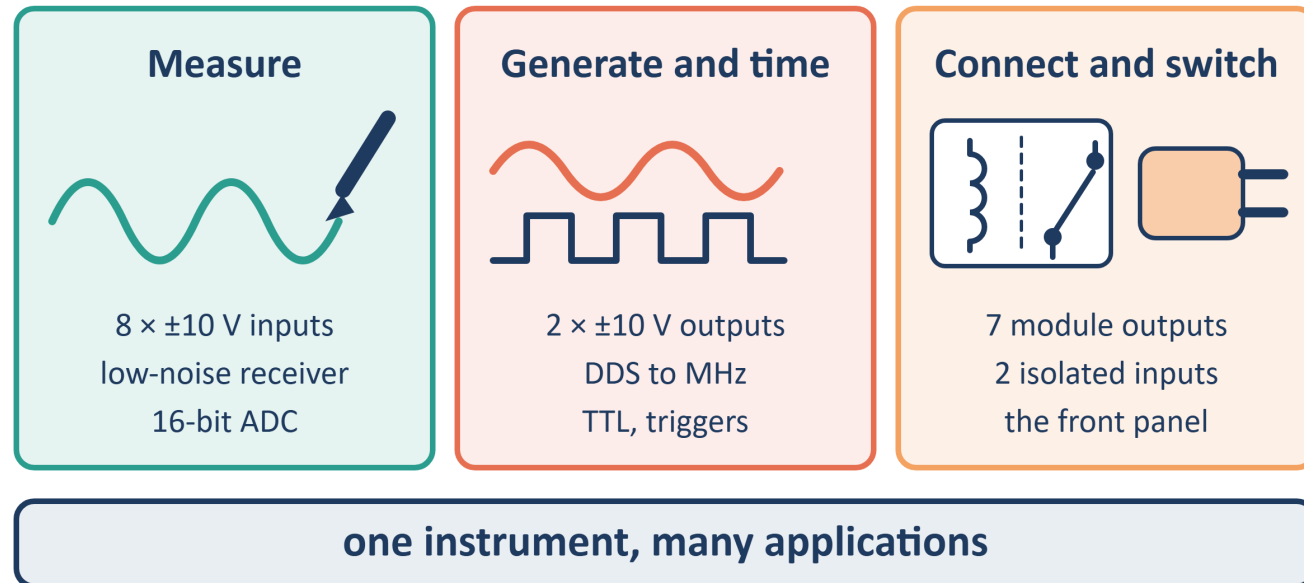


Soldered?

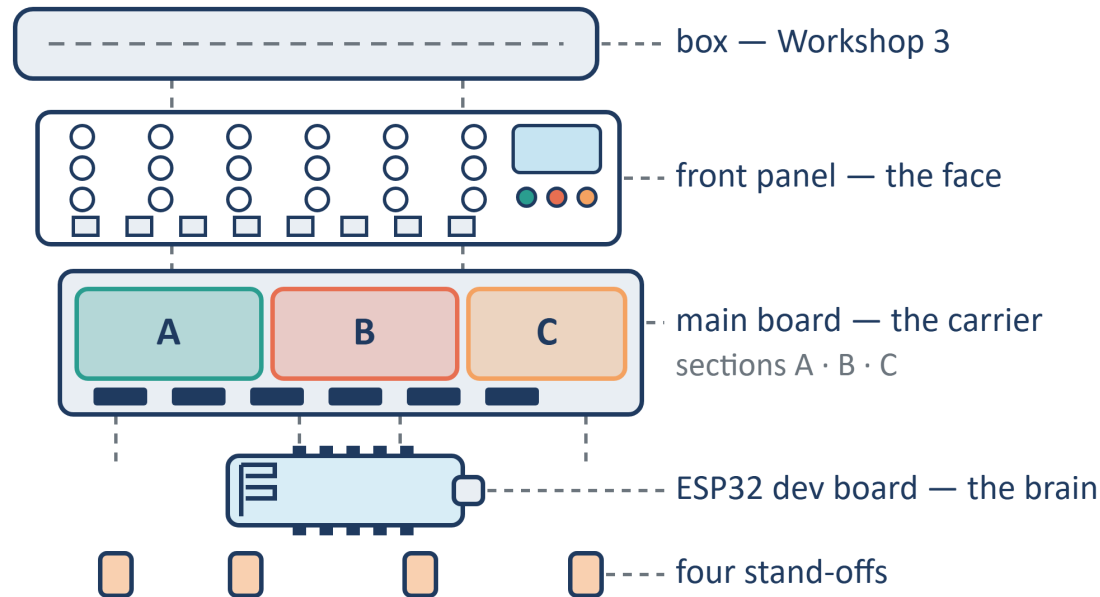


Used CAD?

- Measure: $8 \times \pm 10$ V inputs at 16 bits, and a low-noise receiver
- Generate and time: $2 \times \pm 10$ V outputs, a sine to MHz, TTL, triggers
- Connect and switch: the front panel, 2 optos, 7 module outputs
- Three general capabilities, not three gadgets
- You build one capability; the class builds the instrument



- The face — the front panel: 17 SMAs, a TTL strip, OLED, LEDs
- The carrier — the main board, 180 × 100 mm, Sections A, B, C
- The brain — the ESP32 dev board, plugged into the carrier
- The box — Workshop 3, and it is Project 3b
- Inside: the expansion header, two Qwiic sockets, every jumper



1. NMR console — A, B and C
2. Oscilloscope and waveform generator — A and B
3. Data logger and lab watchdog — A and C
4. Lock-in amplifier and network analyzer — A and B
5. Temperature controller and PID servo — A and C

Every project needs at least two sections

	A measure	B generate and time	C switch and power
NMR console	used	used	used
Oscilloscope + waveform generator	used	used	
Data logger + lab watchdog	used	optional	used
Lock-in + network analyzer	used	used	
Temperature PID servo	used	optional	used

● used ● optional

- Pulse programmer — TTL patterns with microsecond timing
- Motorised stage or shutter controller
- Frequency counter and event timer
- Coil and magnet current controller
- A chat interface to the instrument — LINE or Telegram
- An instrument that explains itself
- Can you adapt it to solve a problem in your research?

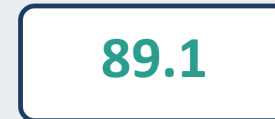
Be creative



pulse programmer



motorised stage



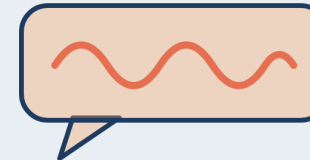
frequency counter



LINE chat



coil current
controller

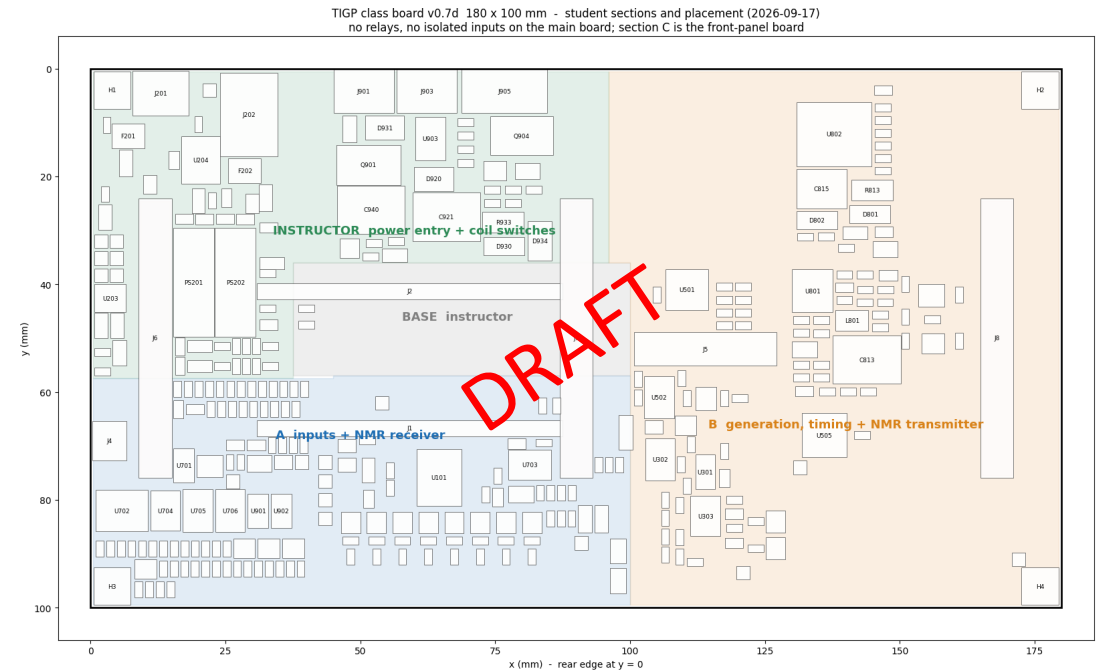


explains itself

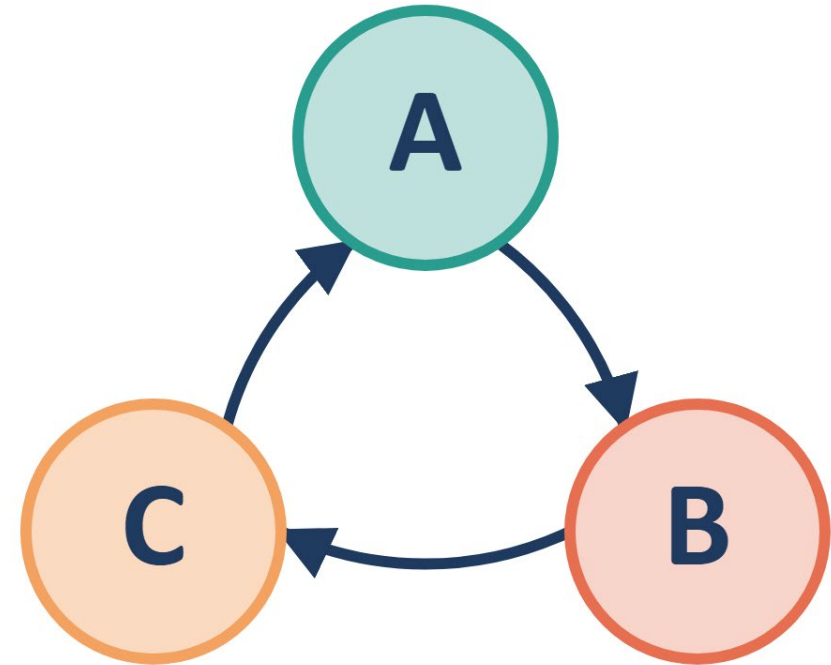
TiGP Three sections, one each



- Section A — the eight inputs and the NMR receiver
- Section B — the outputs, timing and the NMR transmitter
- Section C — the front panel board, optos and module outputs
- The instructor keeps the base, the power entry and the coil switches
- Your zone is the copper you route — stay inside it



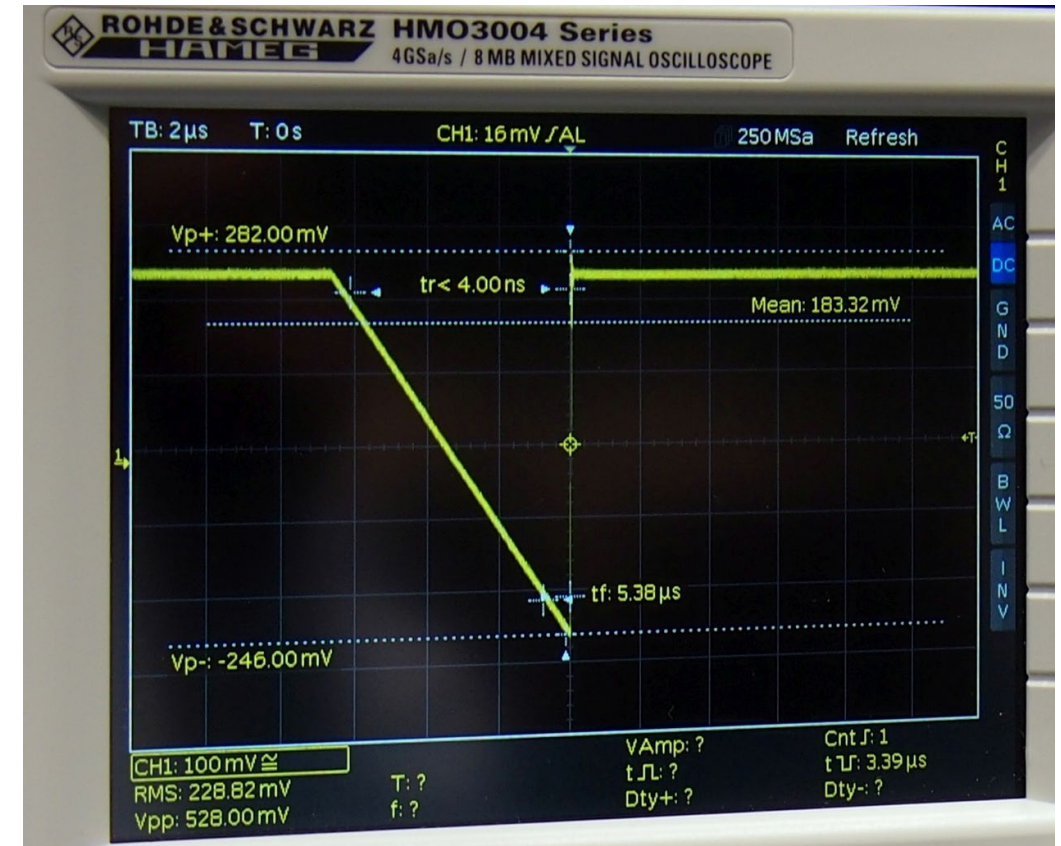
- Say which section you want
- Whatever is left is drawn by lot
- Write your section into your PROGRESS.md — the tutor asks
- Meet your reviewer now: A reviews B, B reviews C, C reviews A
- Everything that enters the board is copper you can explain



A reviews B, B reviews C,
C reviews A

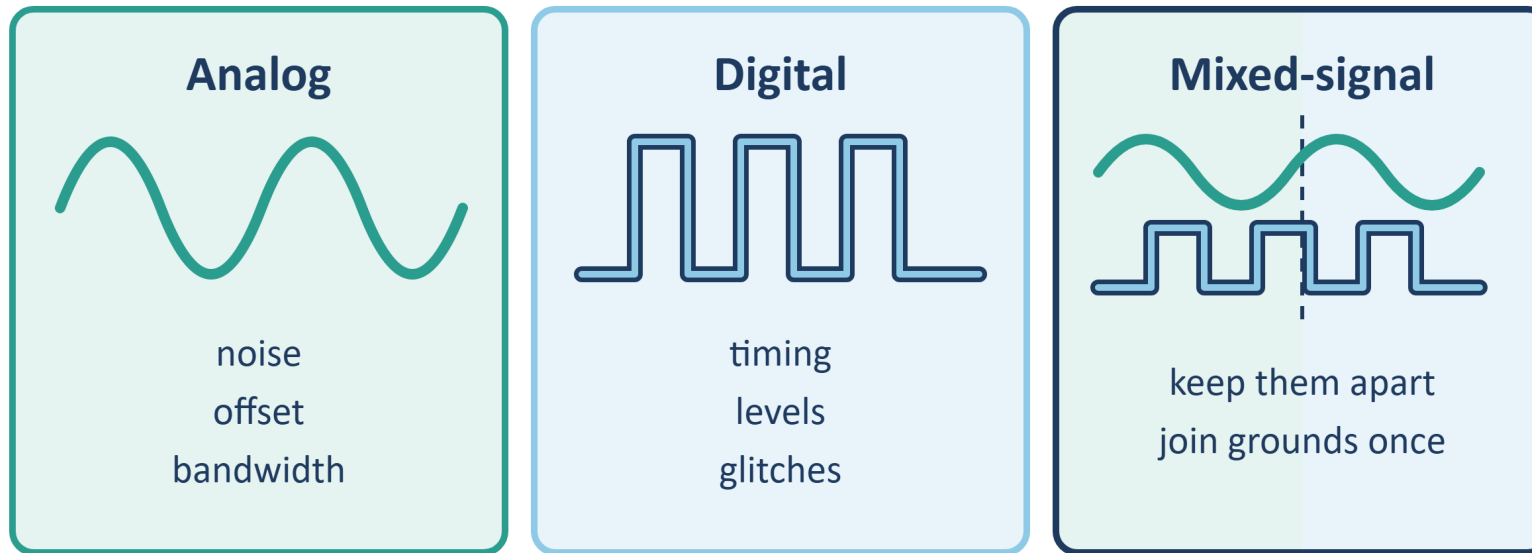
Electronics 101

A crash course to understand what we are building



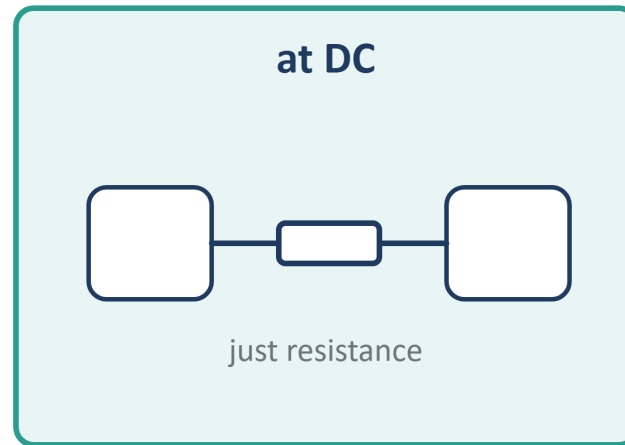
- Analog worries: noise, offset, bandwidth
- Digital worries: timing, levels, glitches
- Mixed-signal: keep them apart, and join the grounds once
- Our board is all three at the same time
- That is exactly what makes the layout the design

three worlds, three worries

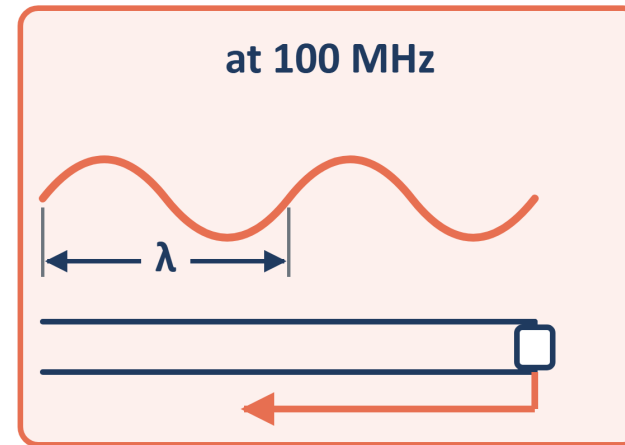


- At DC a wire is a wire, and Ohm's law is enough
- At RF a wire is a transmission line: impedance, reflections
- RF starts when the wire is a fraction of a wavelength

- **Digital signals are RF**, the sharp sigital edges require high frequencies (fourier!)
- 89 kHz on a 180 mm board is not RF
- But a tuned coil and a mixer are RF ideas at any frequency



a wire is a wire



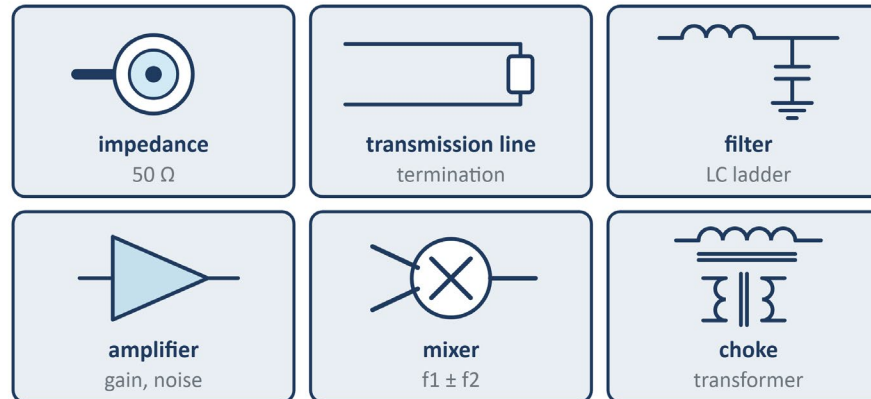
a transmission line

RF starts when the wire is a fraction of a wavelength

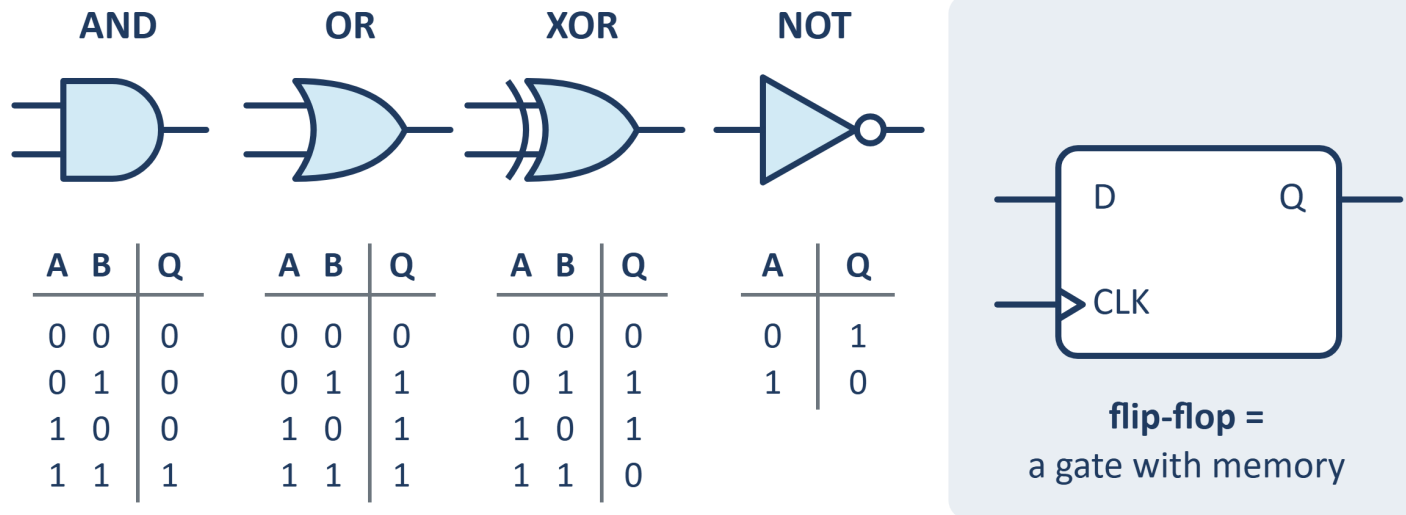
TiGP RF 101 — the vocabulary



- **Impedance:** the ratio of voltage to current a line presents
- **50 Ω :** the standard for cables, connectors and instruments
- **Transmission line:** terminate it, or the edge comes back at you
- **Filter:** passes one band, rejects another — LC, RC or crystal
- **Amplifier:** gain, and a noise figure that says what it added
- **Mixer:** multiplies two signals and gives $f_1 + f_2$ and $f_1 - f_2$
- **Choke and ferrite:** high impedance to noise/RF, low to DC
- **Transformer:** isolation, and an impedance ratio for free
- **Ours:** 49.9 Ω back-termination, ferrites on +3V3A, an I/Q mixer



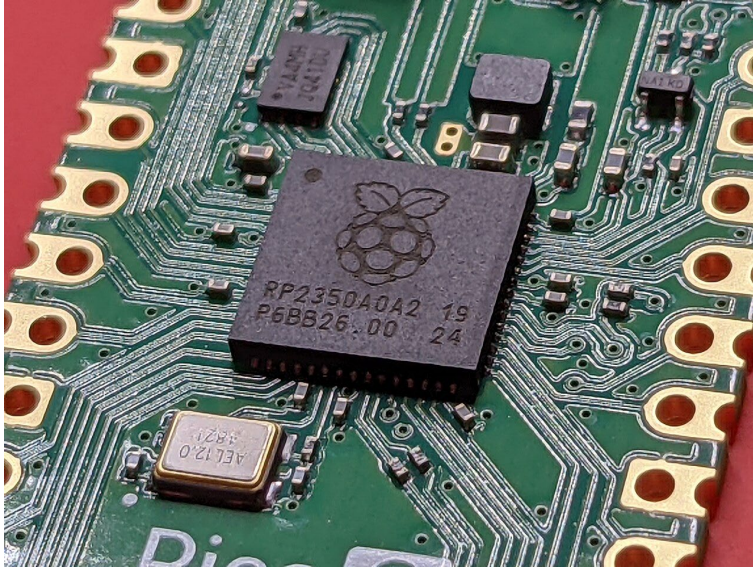
- AND, OR, XOR, NOT — everything digital is made of these
- A flip-flop is a gate with memory: it holds one bit
- A counter is flip-flops in a row
- A processor is a very large pile of the same thing
- Ours: a 74HC74 makes two clocks exactly 90° apart



	74HC	74HCT	74LVC
Supply	2–6 V	4.5–5.5 V	1.65–3.6 V
Input HIGH	$0.7 \times VCC = 3.5 \text{ V}$	2.0 V (TTL)	$0.65 \times VCC$
3.3 V drives it at 5 V?	no	yes	not applicable
Typical delay	about 9 ns	about 10 ns	about 3 ns
On our board	74HC74 divider	74HCT125, AHCT541	74LVC1T45 on TRIG

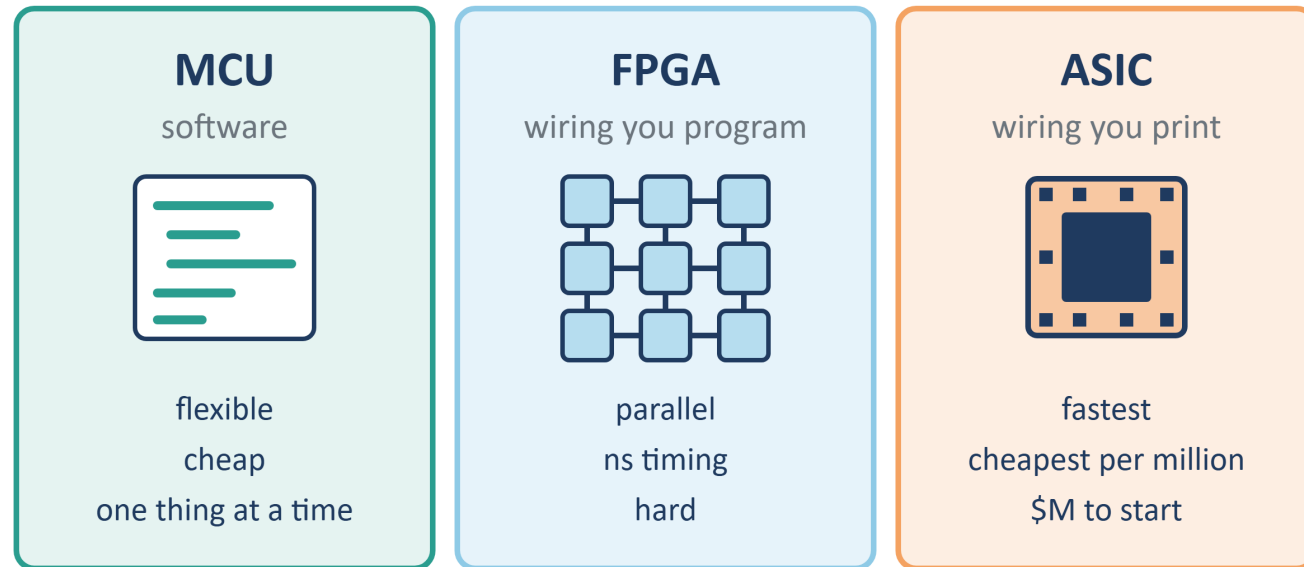
- The family name tells you the supply and the input thresholds
- 74HCT exists precisely to accept 3.3 V and drive 5 V

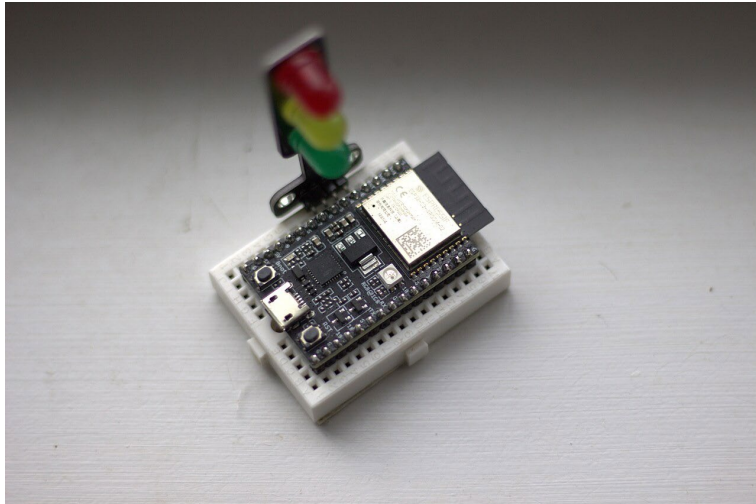
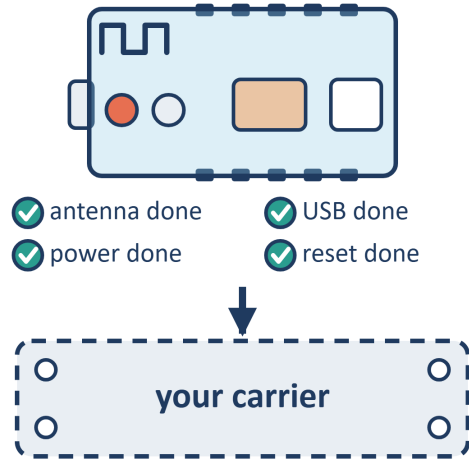
**Never feed 5 V into an ESP32 pin
— it is not 5 V tolerant**



- Microcontroller: one chip — CPU, flash, RAM, ADC, timers, radio
- It runs your bare code and keeps microsecond timing
- Processor: needs RAM, storage, power and an operating system
- An operating system means milliseconds, not microseconds
- Ours: A microcontroller - ESP32-S3, two cores at 240 MHz, 8 MB PSRAM, about US\$10

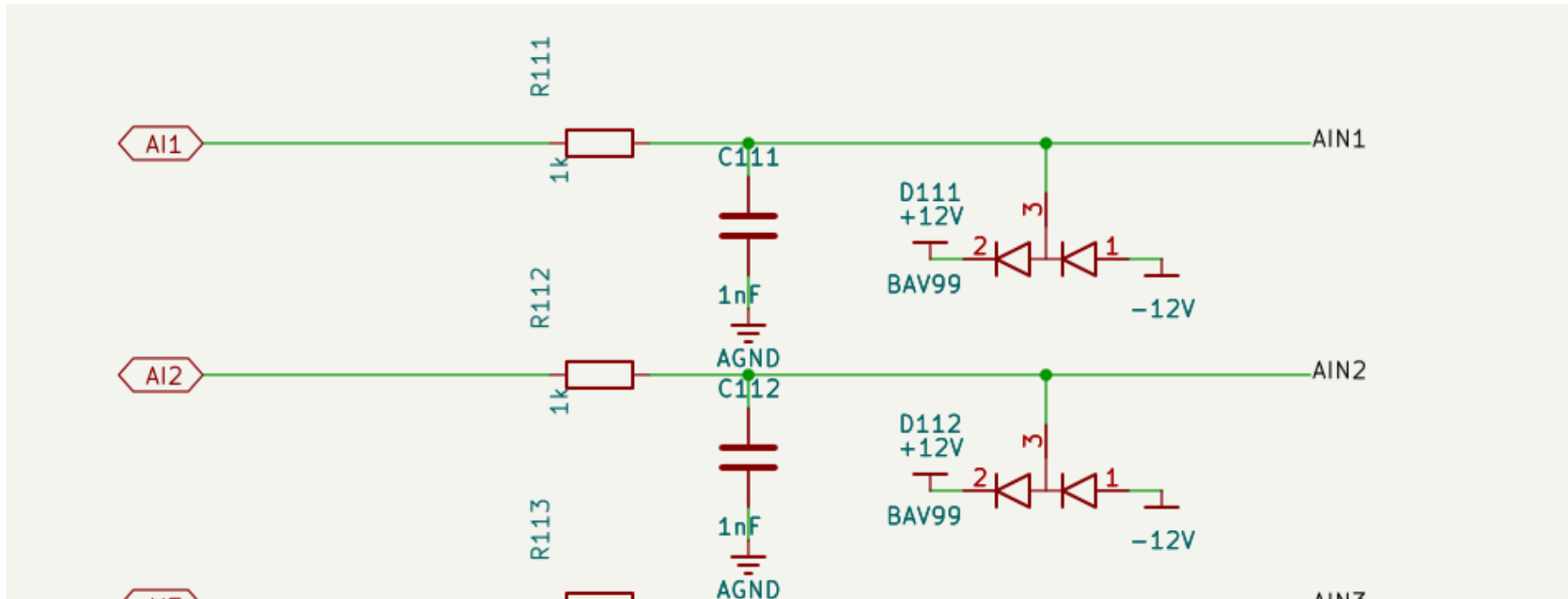
- MCU: software — flexible, cheap, one thing at a time
- FPGA: wiring you program — parallel, nanosecond timing, hard
- ASIC: wiring you print — fastest, and millions to start
- What would need an FPGA here: a 40 MSPS scope channel
- What needs an ASIC here: nothing we do

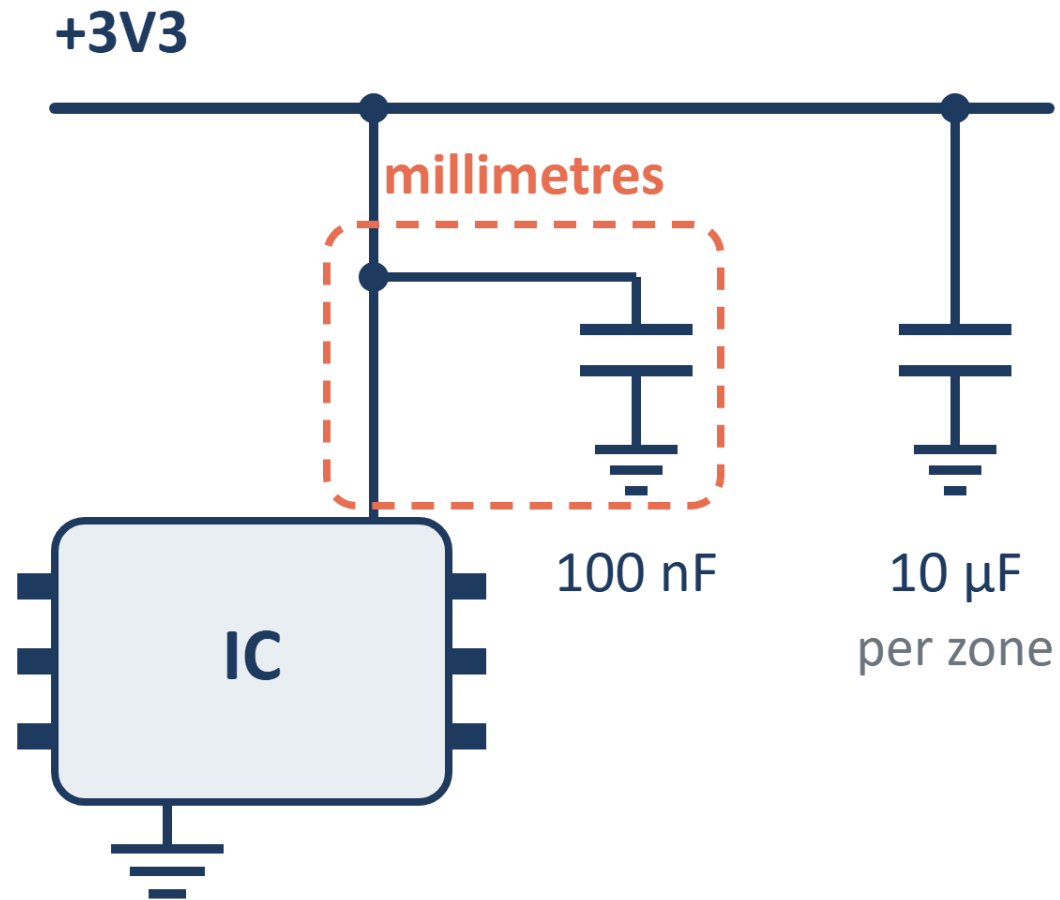




- Power, USB, antenna and reset are already proven on it
- Flash it from VS Code on day one — no board of your own needed
- The bare module stays your fallback if you outgrow it
- Design the carrier, not the CPU
- Ours: a Jinhua ESP32-S3-N16R8 board, socketed on the class board

- Position on the page means nothing at all
- The net name means everything — the same name is the same wire
- A symbol is what a part does; a footprint is where it is soldered
- A label that leaves the sheet reappears on another sheet
- Ours: 10 sheets, 404 parts, names like AI1, +5VA, SPI_SCLK

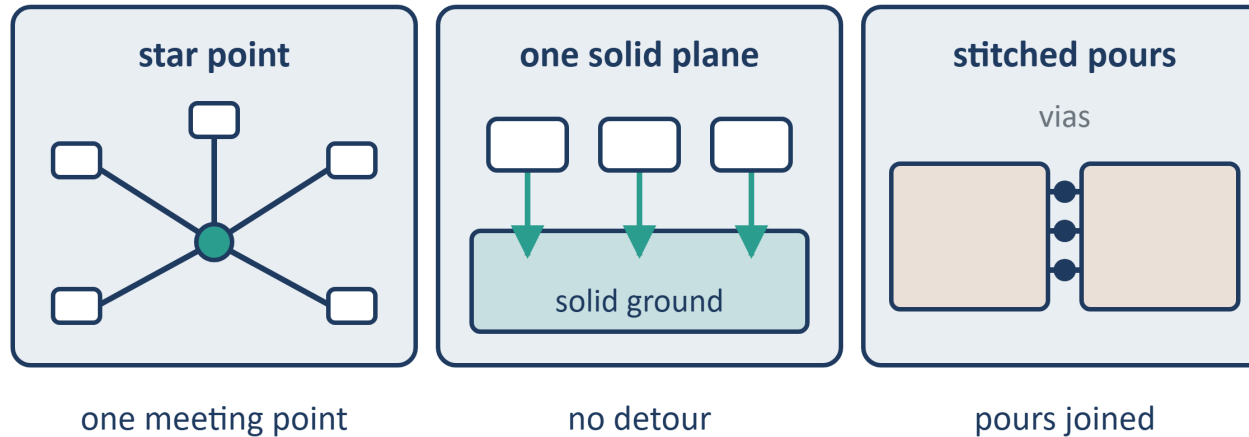




- Every integrated circuit gets 100 nF right at its supply pin
- Every rail in every zone gets a 10 µF as well
- Distance matters more than value: the loop must be millimetres
- The capacitor supplies the nanosecond current the chip draws
- The regulator is far away and much too slow for that
- Every reviewer checks: decoupling next to the pin

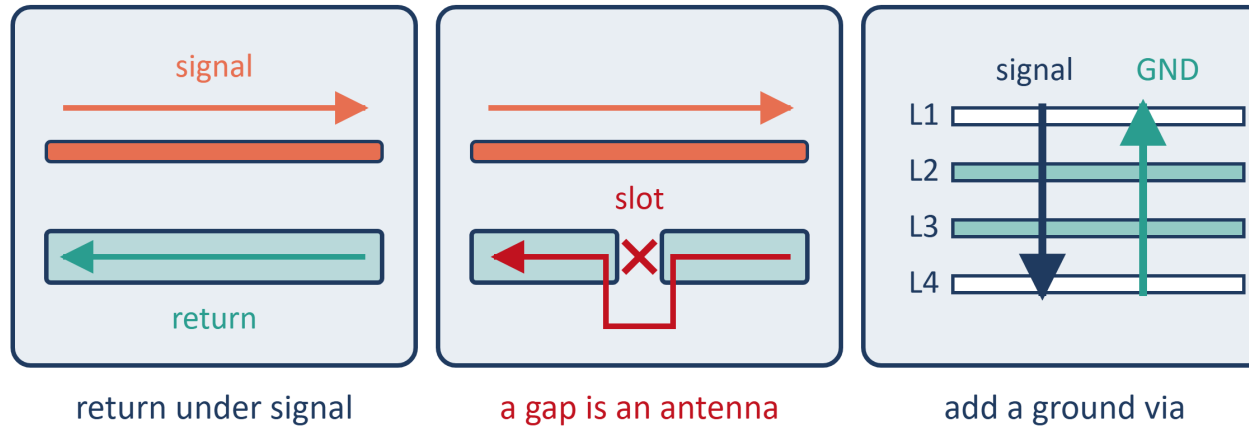
- Current always returns — know where the return flows
- Noisy and quiet returns must not share the same copper
- Star point, one solid plane, stitched pours: three tools
- A solid plane with good placement is the modern default
- Ours: two solid ground planes, an analog pour, one tie

ground is a return path



- The return current runs directly under the signal track
- Give it copper all the way — a gap in the plane is an antenna
- A slot forces the return around it, into a big loop
- Every signal via gets a ground via next to it
- Ground pours are the return path and the shield, not decoration

the return follows the signal

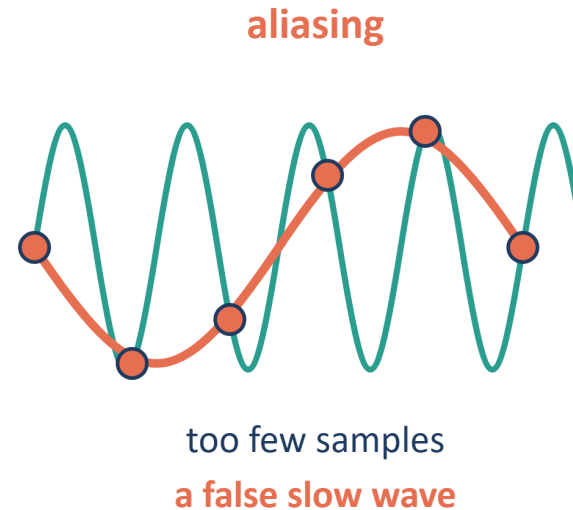
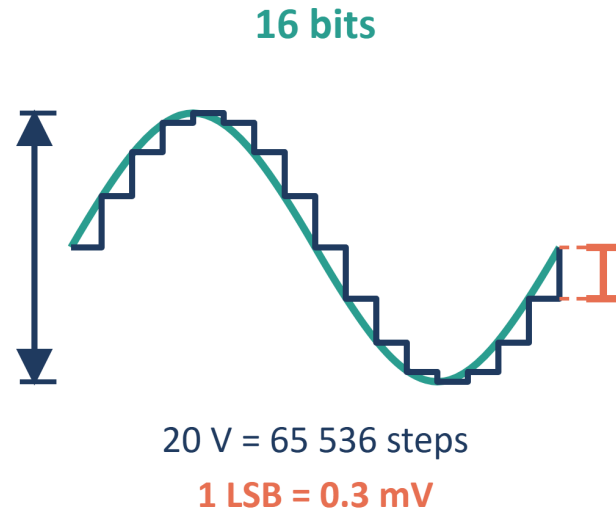


Strategy	Use it when	Failure mode
Star point	low frequency, few blocks	long returns, big loops
Split plane	almost never, in practice	a signal crosses the slot
One solid plane	the modern default	none, if it stays solid
Stitched pours	joining outer copper	starved pours, islands
Via fence	shielding an edge or a line	costs vias and space

- Pick by frequency and by how noisy the loudest return is
- Whatever you pick, never let a quiet return share noisy copper

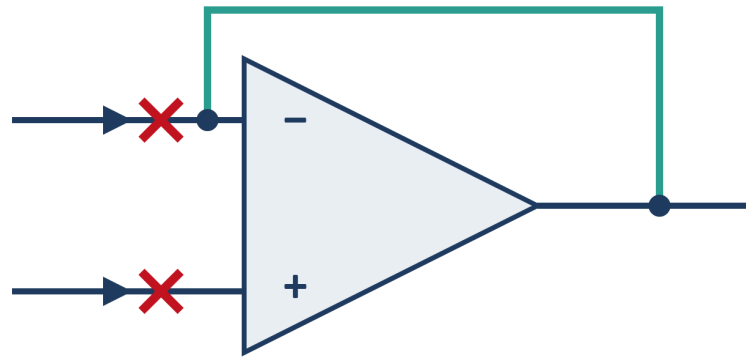
What ± 10 V and 16 bits buy

- 20.48 V across 65 536 steps is 0.3 mV per step
- That step is the unit noise is measured in
- Sample above twice the highest frequency, or it folds back
- Aliasing turns a fast signal into a false slow one
- The whole board exists to keep noise below one step

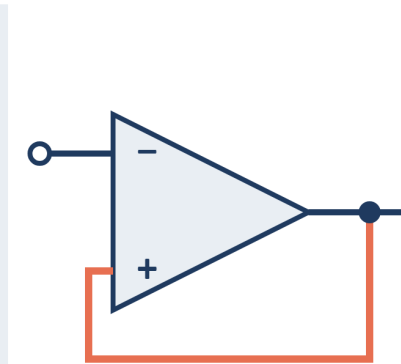


The op-amp's two golden rules

1. The inputs draw no current
2. With negative feedback the output makes the inputs equal
 - Positive feedback runs away instead — on purpose, sometimes
 - That is what a comparator and an oscillator are



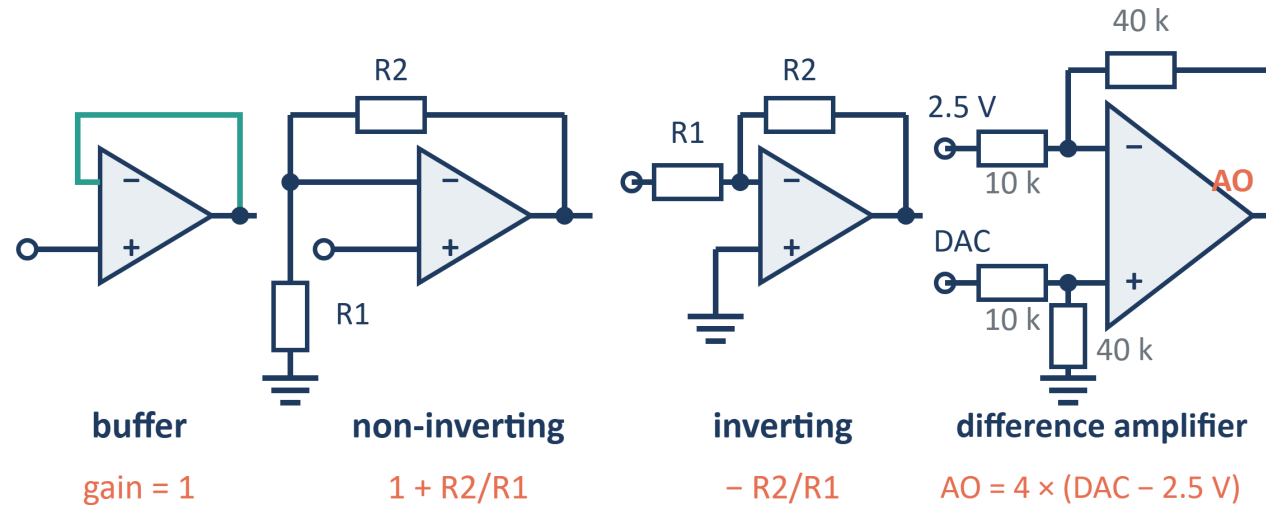
1. the inputs draw no current
2. feedback makes $V_+ = V_-$



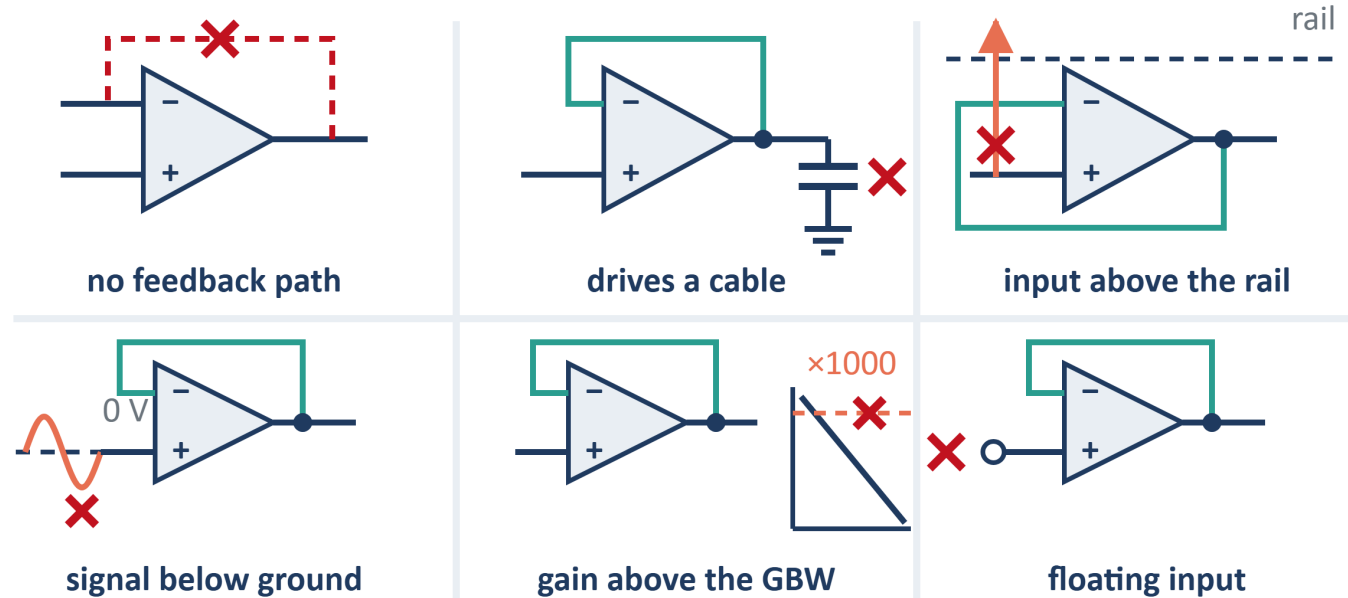
positive feedback
runs away
comparator

Every op-amp circuit on this board
follows from those two rules

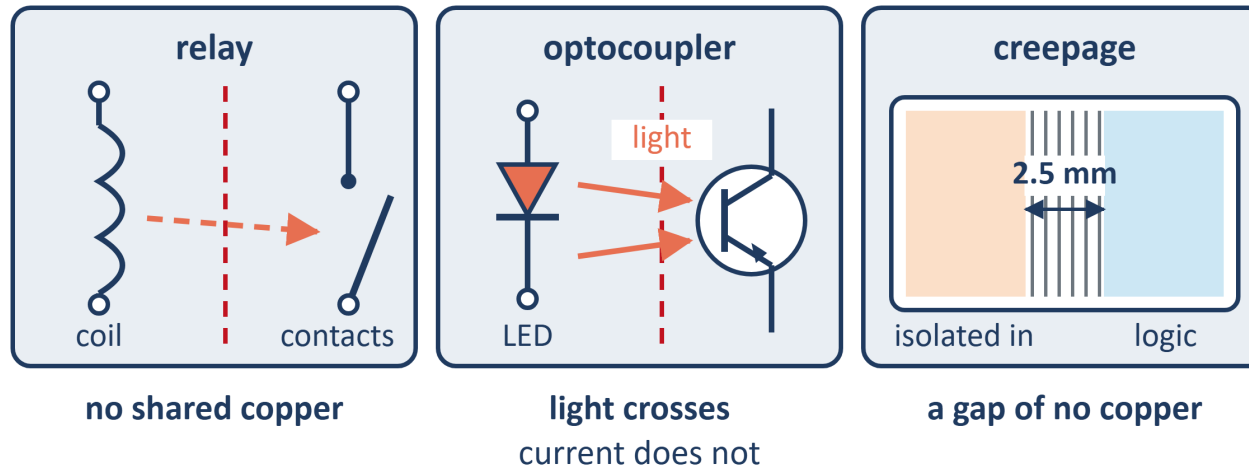
- Buffer: gain 1 — it drives what the source cannot
- Non-inverting: gain $1 + R2/R1$ · inverting: gain $-R2/R1$
- Difference amplifier: ours is $AO = 4 \times (DAC - 2.5 \text{ V})$
- Gain comes from a ratio, so it is as good as your resistors
- The 49.9Ω in series is what protects the output from the cable

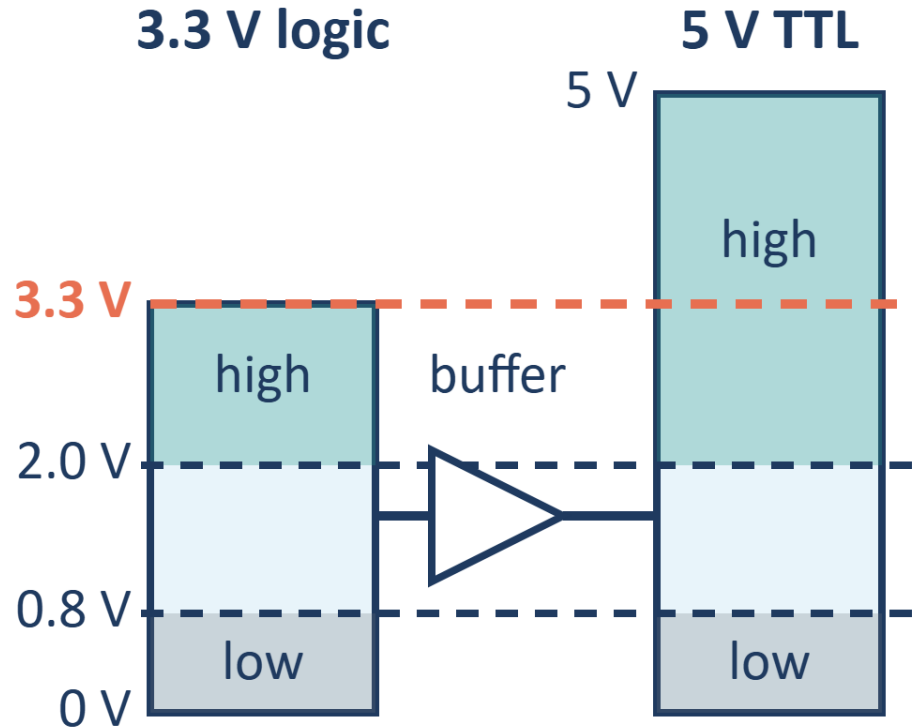


- No feedback path: the output slams to a rail and stays there
- Driving a cable directly: its capacitance makes it oscillate
- Input beyond the rails: phase reversal, or a dead part
- Single supply with a signal below ground: clipped at zero
- Gain above the gain–bandwidth: no gain at your frequency
- Floating input: it amplifies everything in the room



- A relay switches a contact with a magnet — the two sides never touch
- An optocoupler sends light across a gap; current does not cross
- Creepage is the distance that keeps it so
- Copper of any other net under the gap defeats it
- Ours: 2.5 mm at the isolated inputs, and no copper under the band
- Switching beyond the box: 7 module outputs drive isolated modules





3.3 V high is a valid TTL high

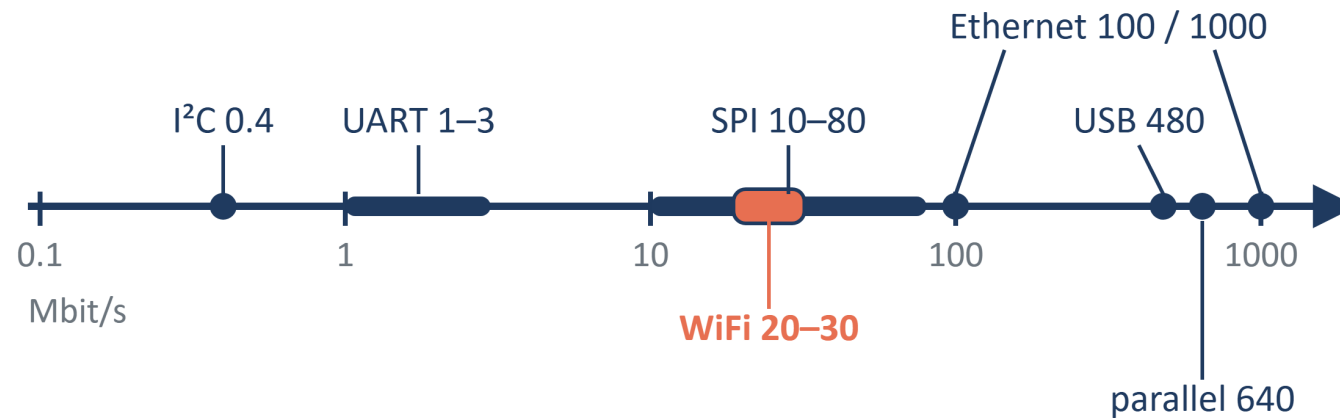
5 V into a 3.3 V pin is not

- The ESP32 outputs 0 or 3.3 V and must never see 5 V on a pin
- 5 V TTL wants a high above 2.0 V, so 3.3 V is a valid high
- Going up is easy; coming down needs a translator or a divider
- A 74HCT part has TTL thresholds: 3.3 V in, clean 5 V out
- “5 V-tolerant” is a datasheet word — our pins are not

Buses: serial or parallel, and how fast

- Serial: few wires, one bit at a time
- Parallel: many wires, one whole word at a time
- I²C 0.4 · UART 1–3 · SPI 10–80 · USB 2.0 480 Mbit/s
- WiFi on the ESP32-S3 gives about 20–30 Mbit/s in practice
- The microcontroller's bus is the instrument's real bottleneck

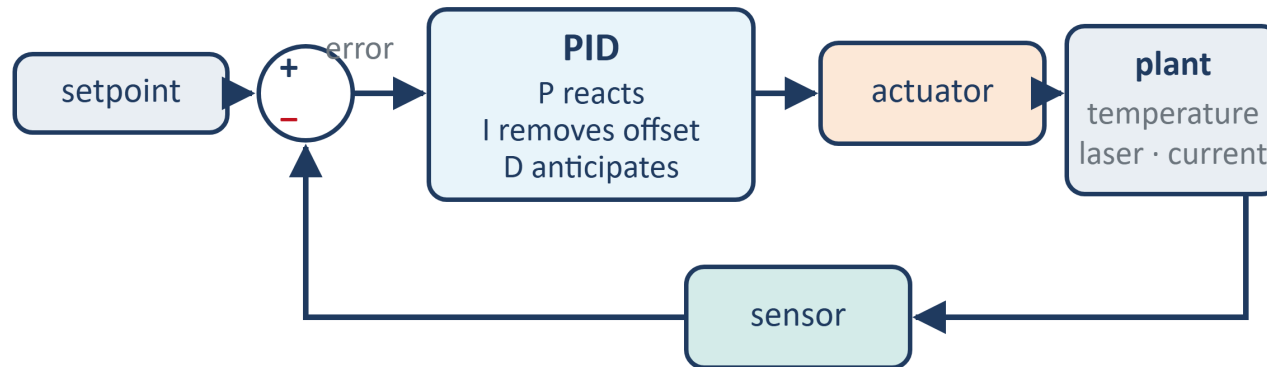
serial: few wires · parallel: many wires



Bus	Wires	On our board	Rate
SPI	clock, data, one select each	ADC, DAC, DDS	up to ~40 Mbit/s
I ² C	two, plus addresses	OLED, expander, clocks	0.4 Mbit/s
USB	one cable	flash and serial	480 Mbit/s
WiFi	none	the phone app	~20–30 Mbit/s
GPIO	one per signal	triggers, gates, blanking	one edge at a time

- Anything timing-critical stays on a direct microcontroller pin
- The I²C expander is why 8 TTL and 7 module outputs cost two pins

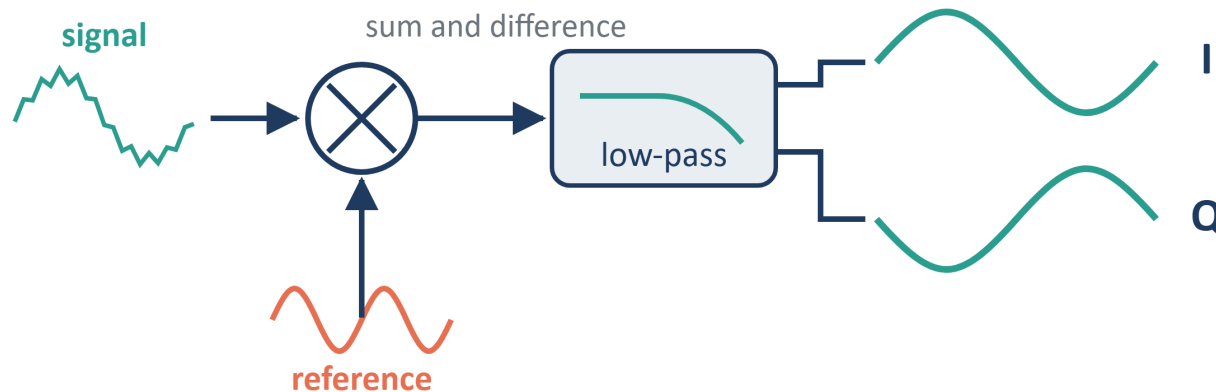
- Measure the thing, subtract the setpoint, act on the error
- P reacts now, I removes the offset, D anticipates
- The same loop locks a laser, a temperature and a coil current
- A loop is only as good as its sensor and its actuator
- Our board has both ends of it, so the loop is firmware



Term	What it does	Too little	Too much
P	reacts to the error now	slow, big error left	rings, then oscillates
I	removes the steady offset	an offset that never goes	slow oscillation, wind-up
D	anticipates the change	overshoot on every step	amplifies sensor noise
Loop rate	how often you correct	sluggish, aliased	noise-limited, wasteful

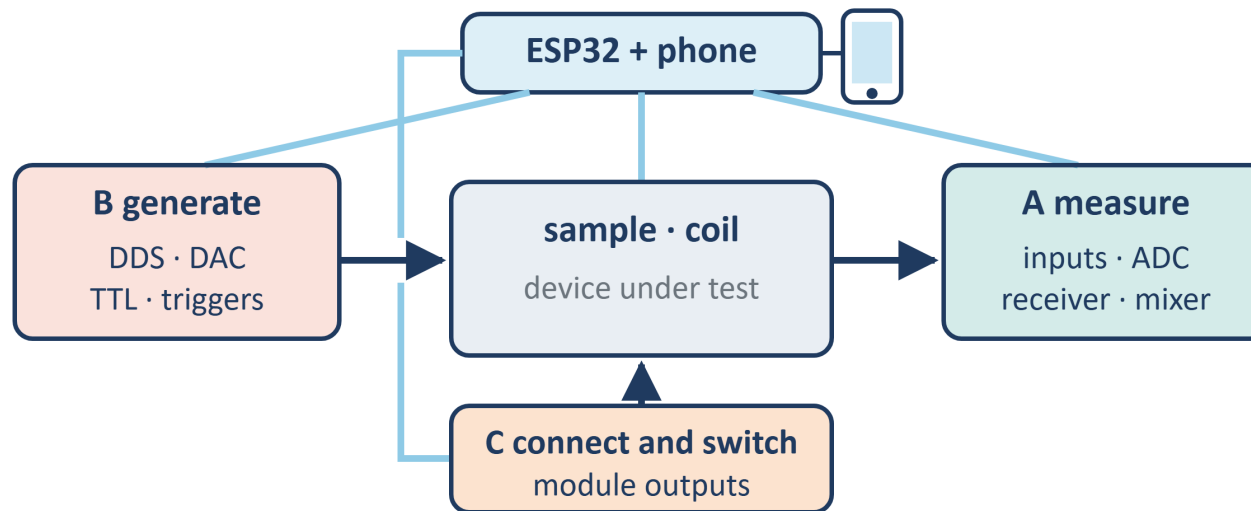
- Tune in this order: P up to ringing, back off, then I, then D
- Watch for integrator wind-up when the actuator saturates

- Multiply the signal by a clean reference at the same frequency
- Keep the difference frequency, filter everything else away
- Noise outside that narrow band simply is not there any more
- Two references 90° apart give I and Q: amplitude and phase
- A lock-in, a radio, an NMR receiver and a network analyzer agree
- Ours: 89.4 kHz down to a 5.4 kHz band the converter can read



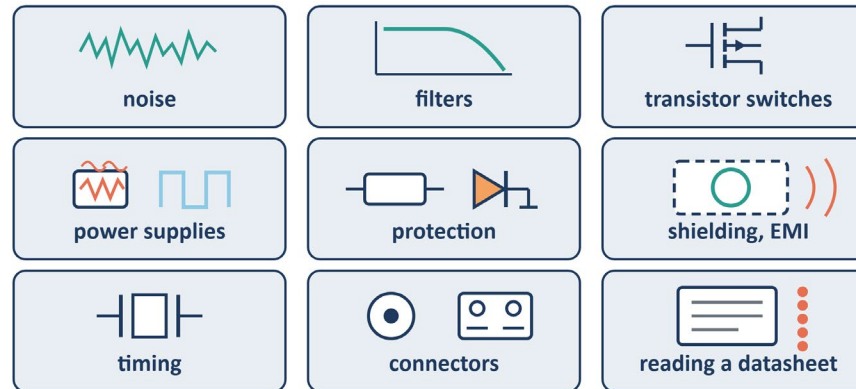
lock-in = radio = NMR = network analyzer

- B generates and times, the sample responds, A measures
- C connects and switches — everything that leaves the box
- The ESP32 and your phone sit on top of all three
- Change the firmware and the same copper is a different instrument
- NMR · scope and generator · logger · lock-in · PID servo

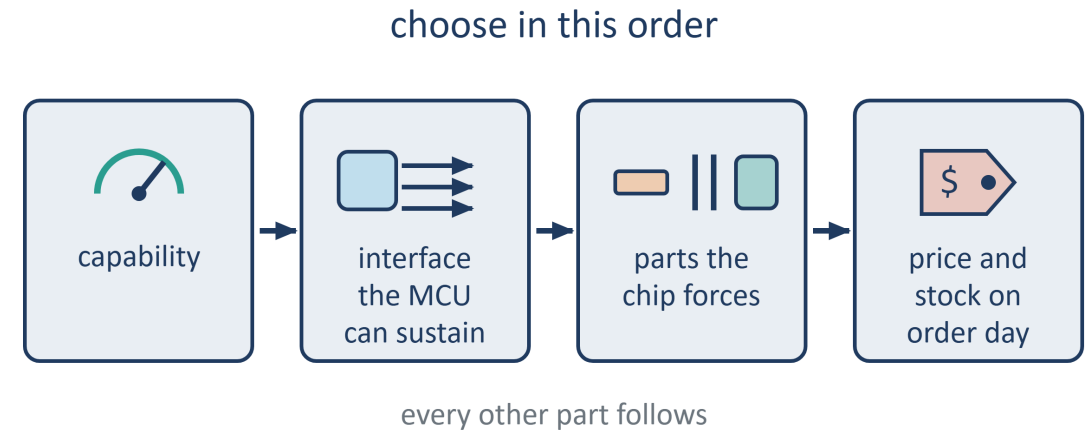


NMR · scope/AWG · logger · lock-in · PID

- **Noise:** Johnson from resistance, shot from current, $1/f$ at low f
- **Filters:** an RC corner at $1/(2\pi RC)$; order sets the steepness
- **Transistors as switches:** MOSFET gate charge, BJT base current
- **Power supplies:** linear is quiet and hot, switching is cool and noisy
- **Protection:** TVS, fuse, clamp diode, reverse-polarity FET
- **Shielding and EMI:** loop area, twisted pairs, ferrites
- **Timing:** crystal or TCXO, parts per million, jitter
- **Connectors:** SMA for signals, screw terminals for power
- **Reading a datasheet:** the five places to look



Choosing chips is the design

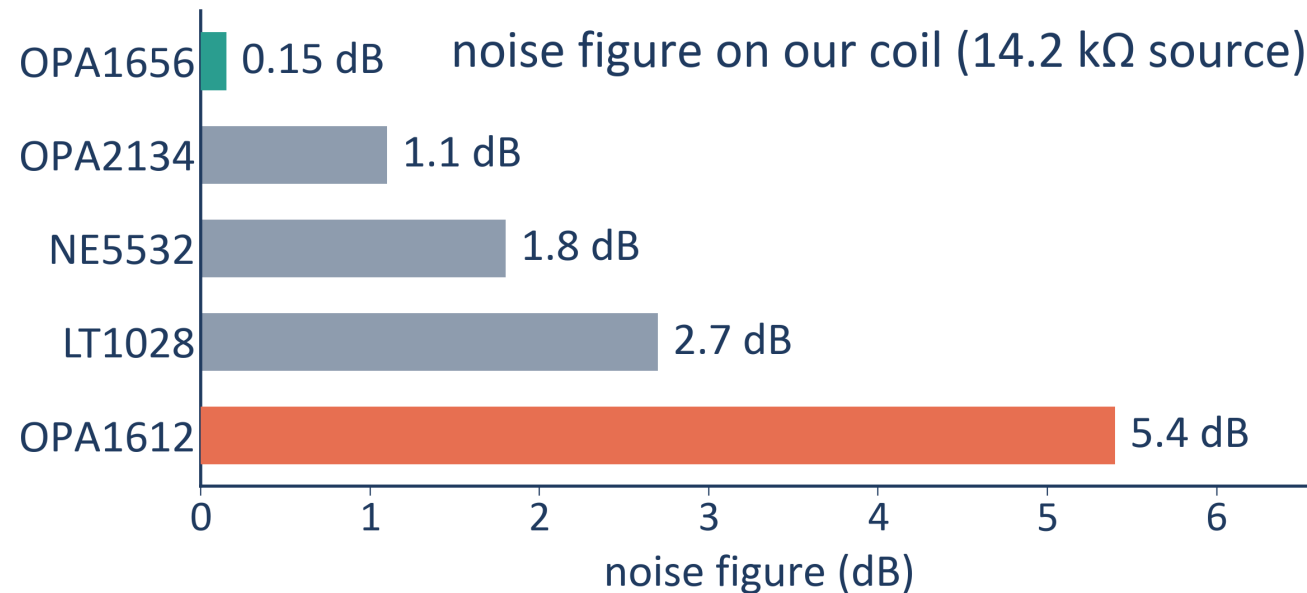


Capability, then the interface the microcontroller can sustain

Then the parts the chip forces you to add, then price and stock

	ADS8688 (ours)	ESP32 ADC	AD7606	ADS9224R	AD9235-40
Bits / channels	16 / 8 muxed	12 / muxed	16 / 8 sim.	16 / 2 sim.	12 / 1
Rate	500 kSPS total	~83 kSPS	200 kSPS/ch	3 MSPS/ch	40 MSPS
Input range	±10.24 V direct	0–3.1 V	±10 V direct	0–5 V diff	~2 V diff
Bus load	16 Mbit/s — fits	internal	~26 Mbit/s	96 Mbit/s — no	480 Mbit/s — no
Price 1 / 10	\$10.88 / 8.79	\$0	\$19.96 / 17.87	\$19.74	\$25.29

- The tuned coil looks like 14.2 k Ω , not 50 Ω
- A high source turns tiny input current noise into big voltage noise
- The coil's own Johnson noise is 15.3 nV/ $\sqrt{\text{Hz}}$ — nobody beats that
- OPA1656 (FET input): 0.15 dB · OPA1612 (bipolar): 5.4 dB
- 5.4 dB costs 3.5 times as many averages for the same result
- Below ~ 1 k Ω use bipolar; above ~ 10 k Ω use FET



	DAC8563 (ours)	PWM + filter	MCP4822	DAC8830	Parallel DAC
Bits / channels	16 bit / 2	~8–10 bit (est.)	12 bit / 2	16 bit / 1	8–14 bit / 1
Reference	internal 2.5 V	the 3.3 V rail	internal 2.048 V	external, buy one	external
Output	buffered voltage	filtered square	buffered voltage	unbuffered	current
Extra parts	74HCT125, already on	one RC per channel	none	ref + 2 op-amps	ref + amp + pins
Price / stock	\$7.78 / 2 357	\$0	\$2.73 / 4 279	\$10.29 / 3 272	out of stock

	AD9834 (ours)	AD9833	AD9851	AD9959	Si5351A
Clock / max out	50 MHz / ~20 MHz	25 MHz / 12.5 MHz	180 MHz / 70 MHz	500 MHz / 200 MHz	— / 200 MHz
Tuning step	0.186 Hz	0.093 Hz	32 bit	32 bit	fractional PLL
Phase control	2 registers + a pin	registers only	5-bit phase	per channel, 4 ch	none below 4.7 MHz
Output	sine	sine	sine	sine	square only
Price / stock	\$11.70 / 824	\$13.51 / 1 027	\$45.41 / 5 pcs	\$99.44 / 0	\$1.20 / 35 858

From a schematic to a soldered board

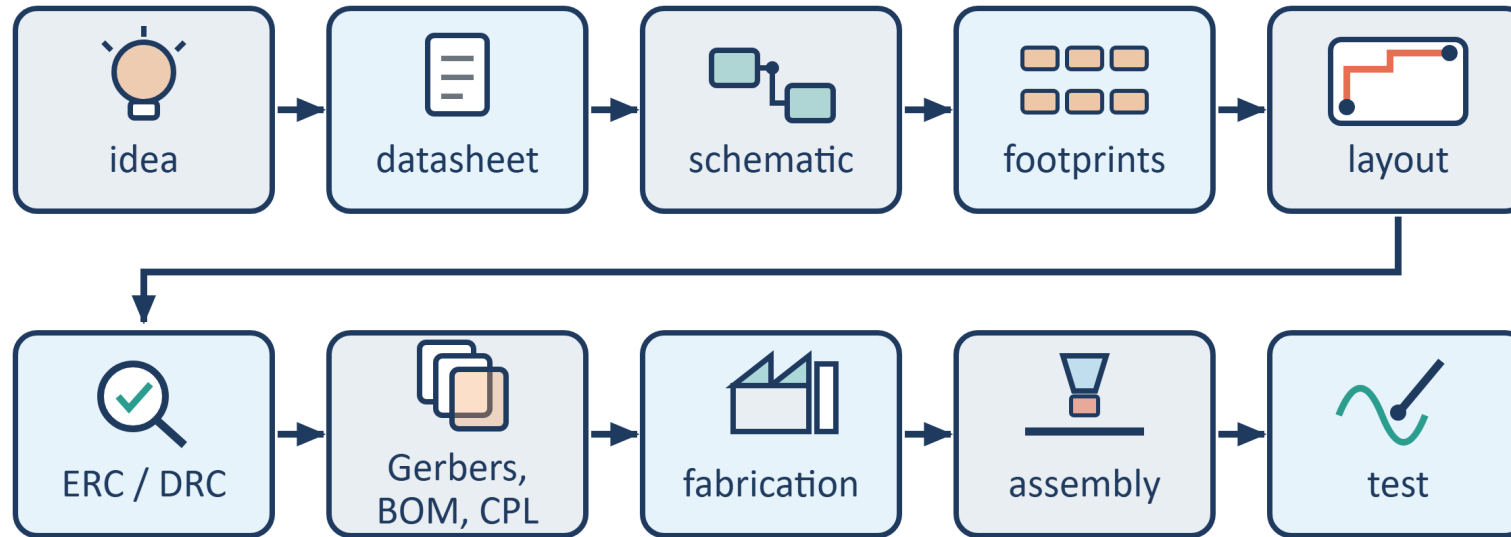
KiCad → Gerbers, BOM, CPL → the factory → an assembled board



TiGP The pipeline, end to end



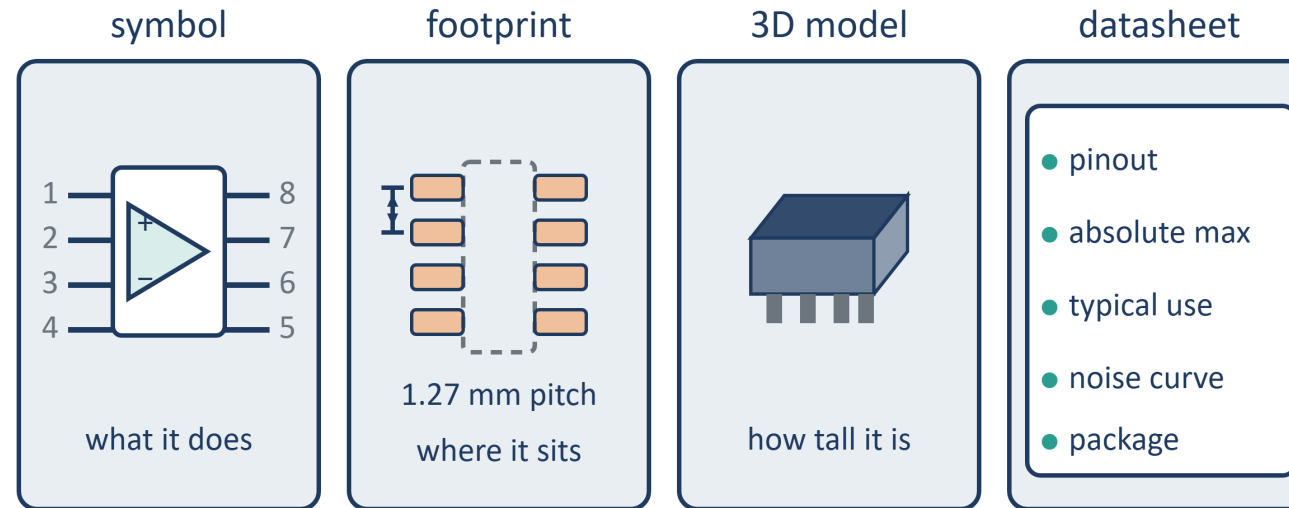
- Idea → datasheet → schematic → footprints → layout
- → ERC and DRC → Gerbers, BOM, CPL → fabrication → assembly → test
- Every step has a checker — run it before you move on
- Today you work in the middle: schematic gaps, then copper



TiGP One part, four views

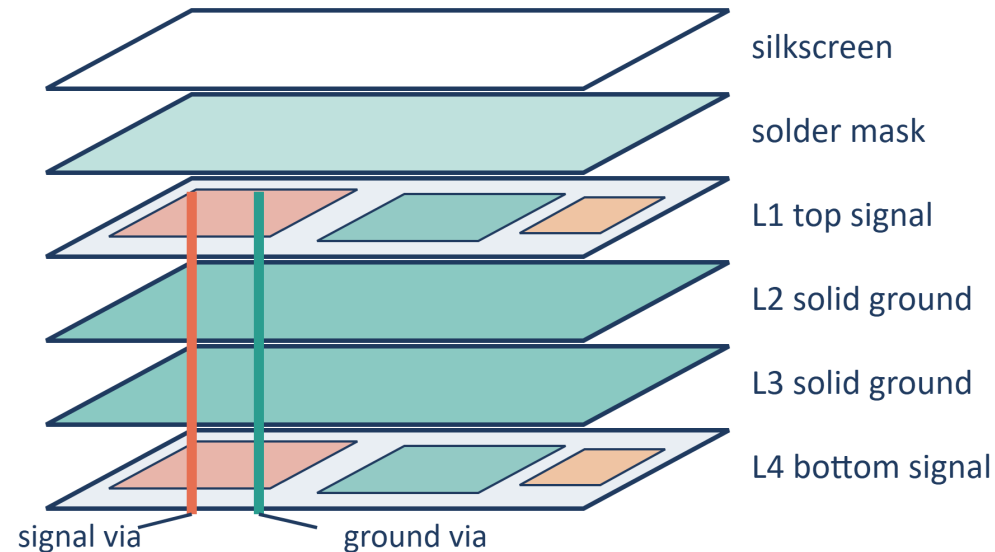


- Symbol: what the part does, and which pin is which
- Footprint: the copper it lands on — pads, spacing, pin 1
- 3D model: what it looks like, and whether it fits the box
- Datasheet: the contract — pinout, limits, the typical circuit
- A right symbol with a wrong footprint is a board you throw away



one part, four views

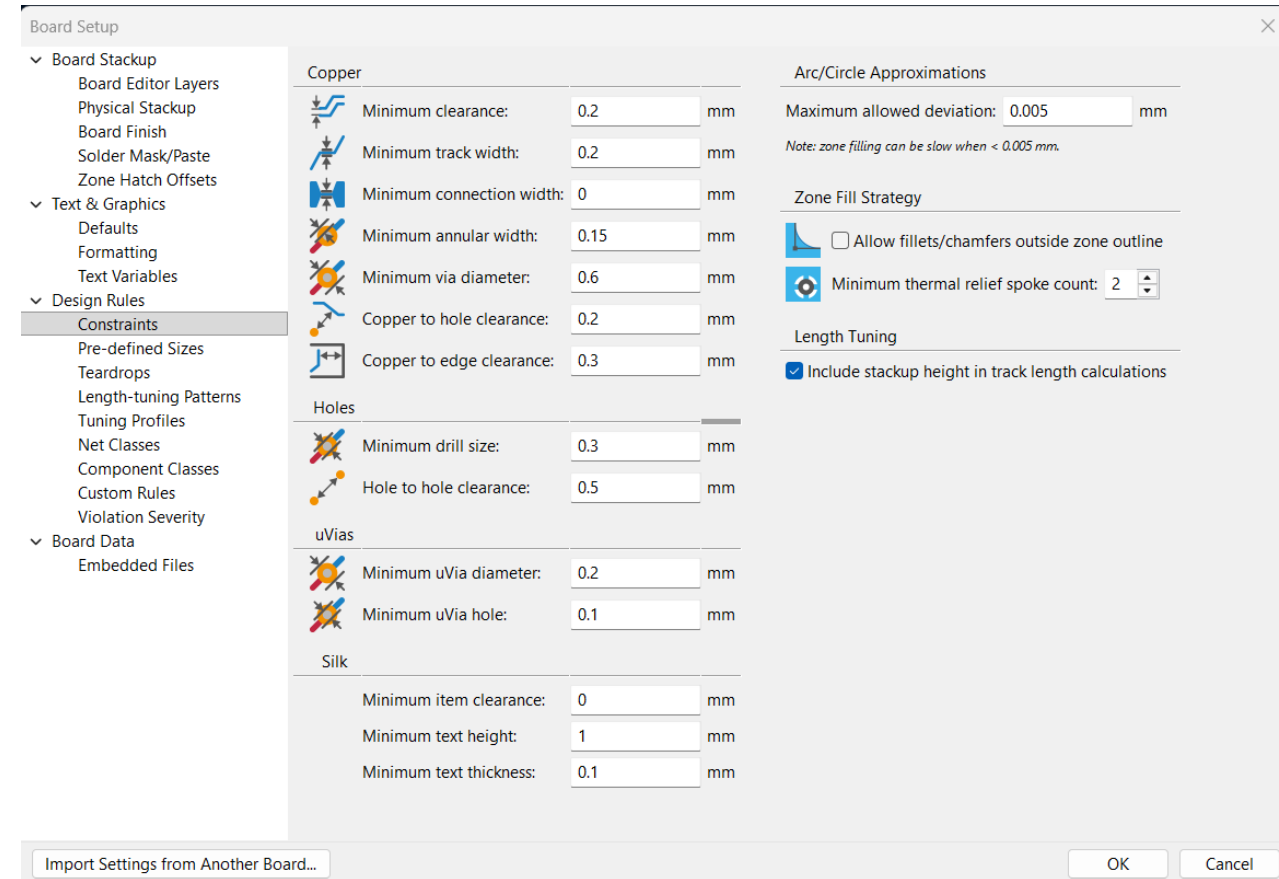
- Copper, insulator, copper, insulator, copper — then mask and ink
- A via is a plated hole that connects one layer to another
- Ours: 4 layers, JLC04161H-7628 stack-up, 1.6 mm thick
- Both inner layers are solid ground — no tracks there at all
- So every outer track has a ground plane right beneath it
- The green is solder mask; the white is silkscreen



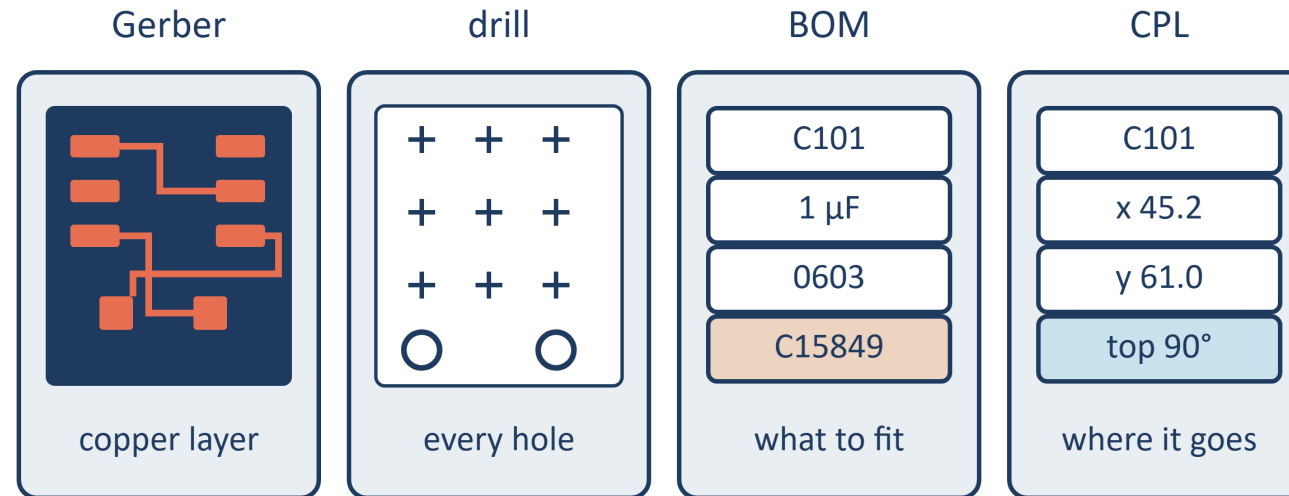
	2 layer	4 layer (ours)	6 layer
Ground	a pour with gaps	two solid planes	planes plus signal
Cost	cheapest	about 2 × a 2-layer	for dense boards
Use it for	simple, slow boards	mixed-signal — our case	BGA, fast buses
Vias	through only	through, 0.3 mm drill	blind and buried
Clearance	0.2 mm signal	2.5 mm at the optos	the same rules

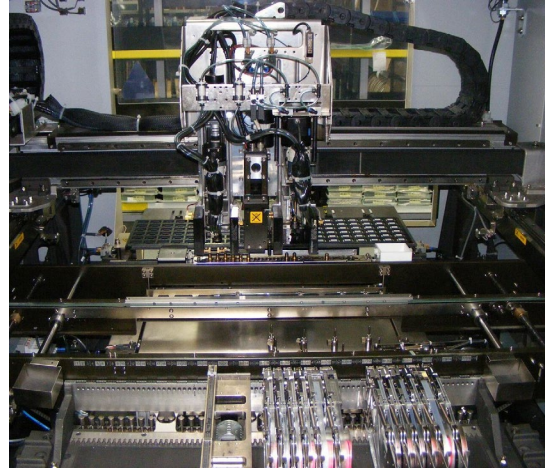
- More layers buy quiet references and routing room, not speed
- Track width follows current; clearance follows voltage

- Minimum track width, clearance, drill size, annular ring
- The factory sets them; KiCad checks them for you
- 0 errors on the design rules check is the entry ticket
- It is not proof of a good design — only of a makeable one
- The electrical rules check is the same idea for the schematic
- Our extra rules: no inner-layer tracks, 2.5 mm at the optos



- Gerbers: one file per layer, the copper as a picture
- Drill file: where every hole goes and how big it is
- BOM: which part goes at which reference designator
- CPL: x, y, side and rotation for every part to be placed
- A part without an LCSC number is a part nobody will solder
- Check the rotations in the factory's upload preview



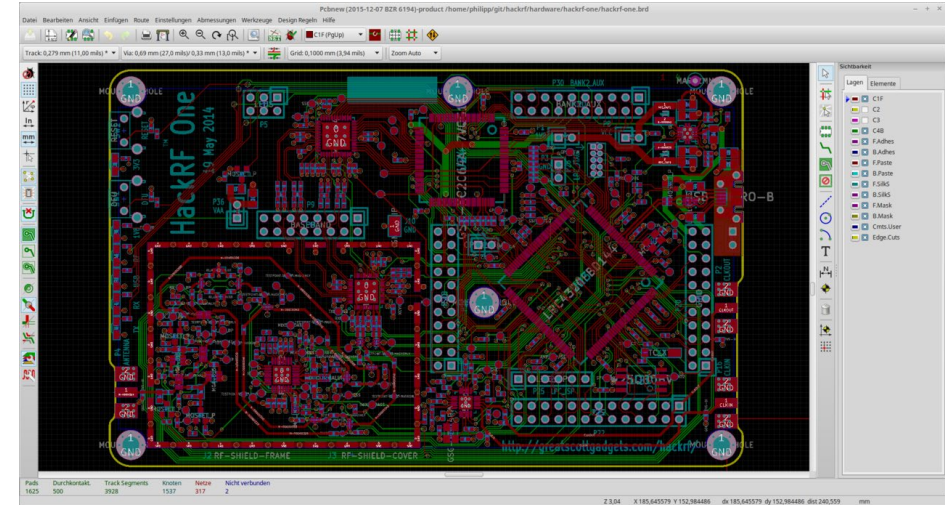


- Surface mount: solder paste through a stencil, place, bake
- The oven's heat profile melts the paste and pulls parts straight
- Through hole: leads through holes, soldered from underneath
- JLCPCB does both — you solder nothing
- The instructor hand-solders the six headers to the panel

Item	Number	Cost
Main boards, 4 layer	7 (5 assembled)	—
Front panels, 4 layer	6	—
Distinct part lines	about 99	about US\$65 / board
Extended lines	about 60 × US\$3	about US\$180
The whole order	—	about US\$720

- Bare boards are cheap; parts and per-line fees are not
- Every Extended part line adds a loading fee to the order

- Place a part (A), wire it (W), set its value (V) and footprint (E)
- Annotate, so every part has a unique reference designator
- Tools → Update PCB from Schematic
- Drag the footprints to where they belong
- Route with X; change layer with a via, V
- Inspect → DRC — then Alt+3 for the 3D view



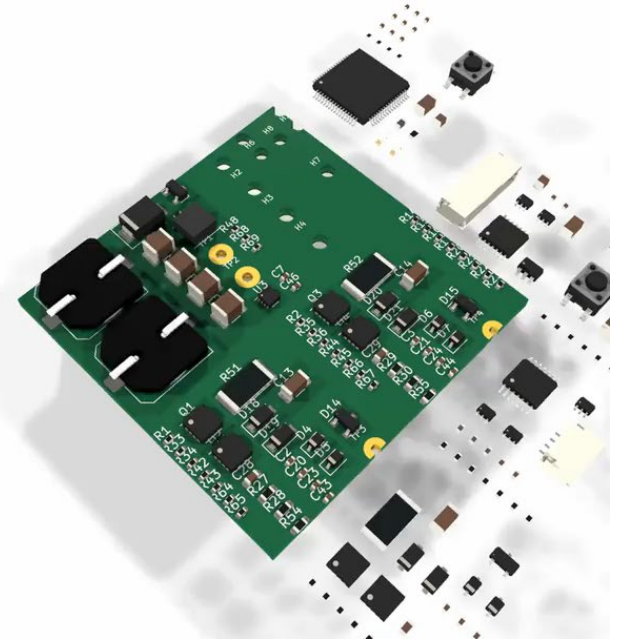
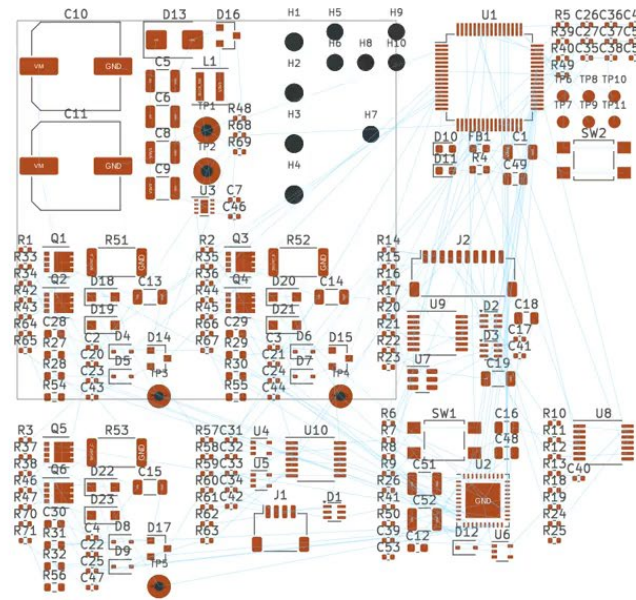
SCREENSHOT — Six KiCad screenshots in a 3 × 2 grid, all on the class project: (1) placing a symbol with the Add Symbol dialog open, (2) the Annotate dialog, (3) Tools → Update PCB from Schematic with its result window, (4) dragging a footprint in the PCB editor, (5) routing a track with X, ratsnest visible, (6) Inspect → DRC showing the violations list.

- Search the factory's own parts library before you draw anything
- Basic parts are free to place; Extended cost a fee per line
- Ours: about 99 lines, roughly 60 Extended at about US\$3 each
- The LCSC field in KiCad is what ties a symbol to a real part
- The LCSC suite plugin imports symbol, footprint and 3D model
- Then check that footprint against the datasheet — it can be wrong



SCREENSHOT — Two windows side by side: a KiCad symbol Properties dialog with the LCSC field visible (for example C1849431 on the OPA1656), and the LCSC suite plugin's search window showing the same part with stock, price and its Basic/Extended status.

The coming wave: AI in PCB design



A four-line brief to a routed two-layer board,
in under three minutes

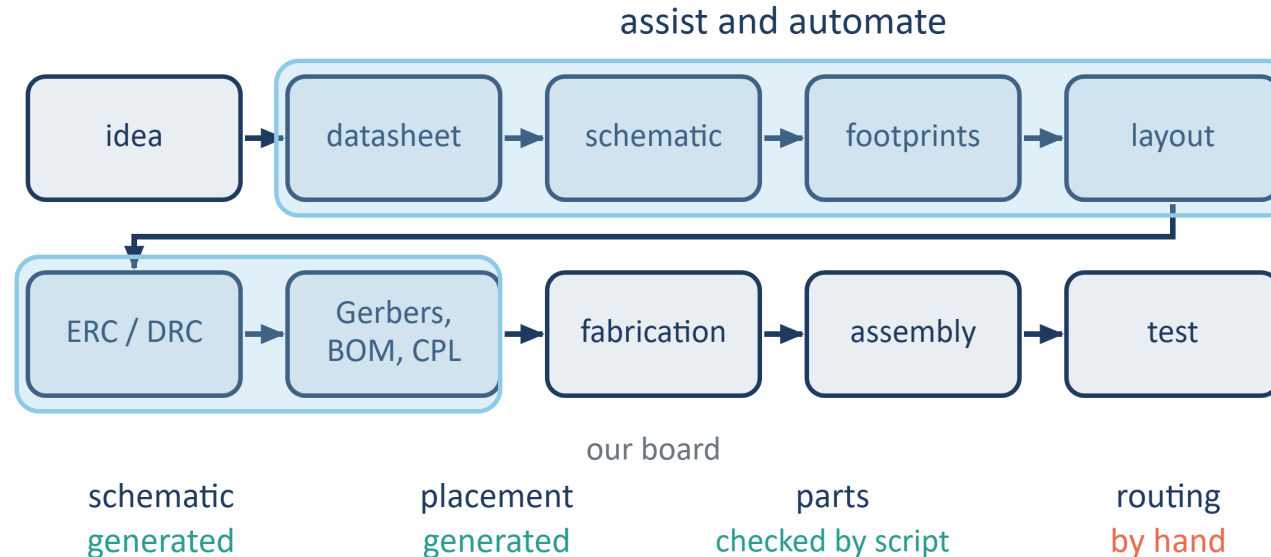
A two-sheet schematic and a 44 × 34 mm
board

<https://openai.com/index/gpt-6-astra/>

- MCP is a plug that gives an AI hands inside a program
- Konnect is a KiCad 10 plugin: schematic, layout, checks, export
- Freerouting and KiCad Routing Tools route; kicad-happy reviews
- The LCSC suite finds and imports parts from inside KiCad
- All free and open source, months old, changing weekly
- SETUP.md Step 10 — optional; the tutor installs it with you

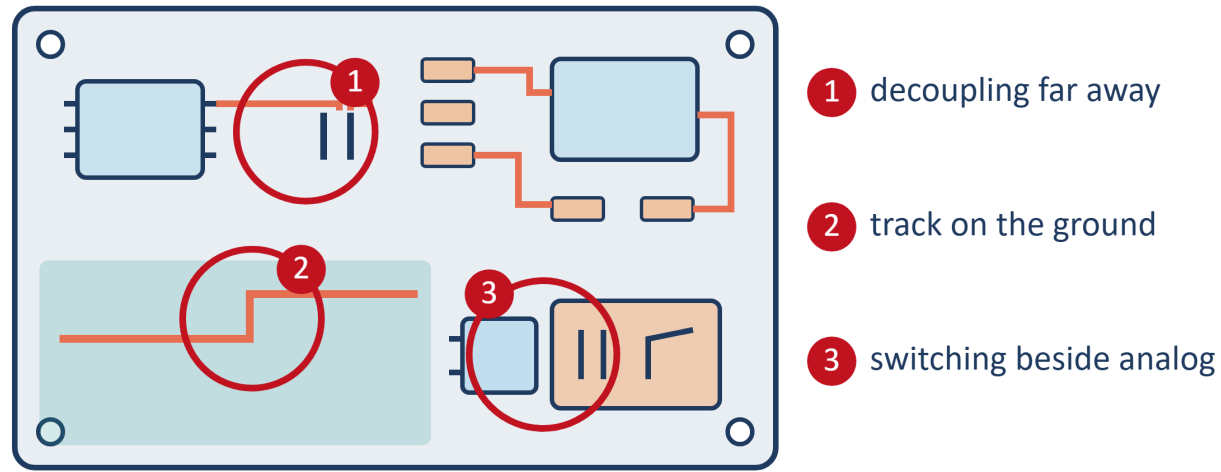
SCREENSHOT — VS Code with a Claude Code session on the left and KiCad open on the class board on the right, mid-task, with a Konnect tool call and its result visible in the Claude transcript (for example listing the symbols in the open schematic, or running DRC).

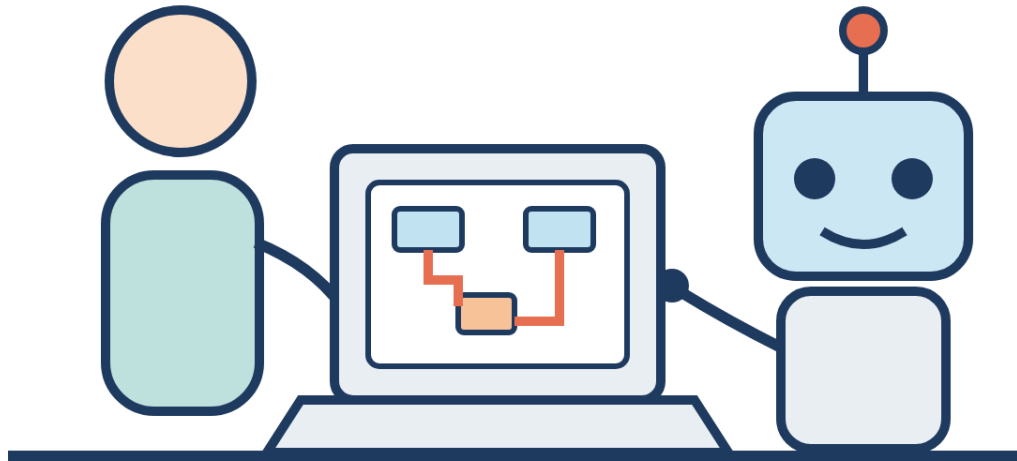
- It reaches: datasheets, schematic, parts, placement, checks, export
- Our board: schematic generated, placement generated
- Parts checked by script; routing done by hand
- Routing is by hand here because microvolts sit beside amps
- An autorouter routes a placement; it does not know why
- Assist and automate — not replace



- The AI's mistakes look exactly like finished work
- Decoupling 15 mm from the pin, on a board that looks routed
- A track on a layer that is supposed to be solid ground
- A switching line running beside the receiver
- You find them only if you know what right looks like
- Check by mechanism: ERC, DRC, the checklist, a fresh session

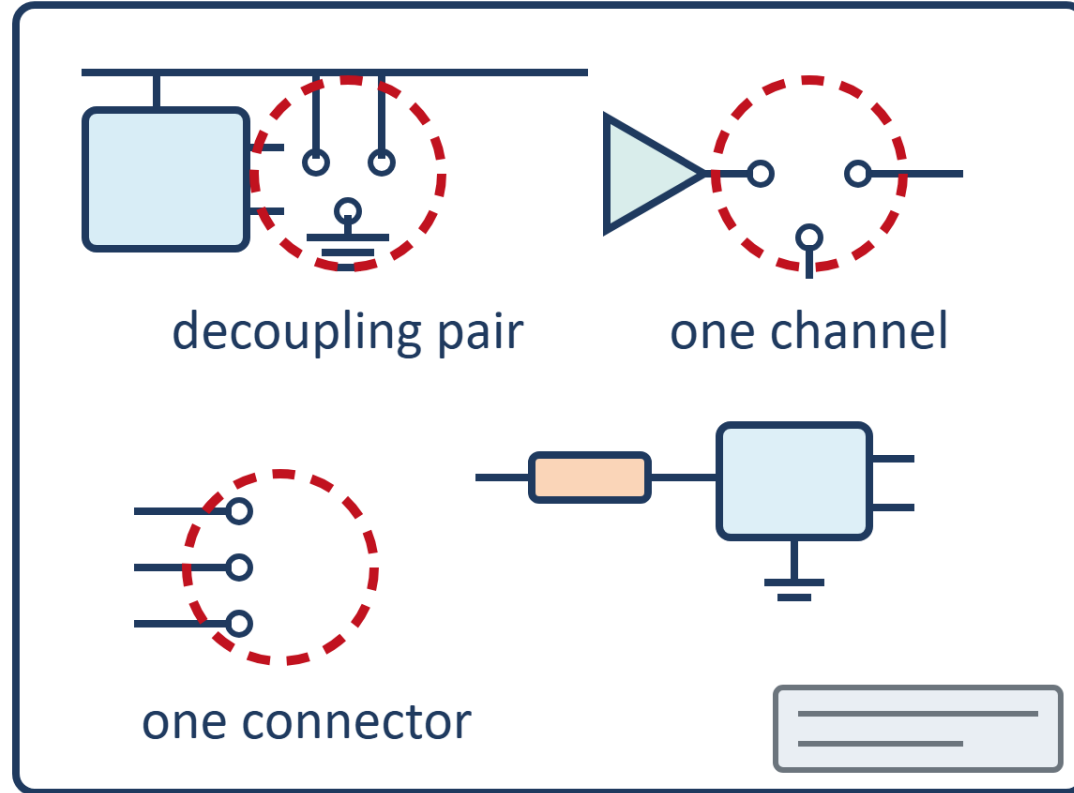
it looks finished





play — then check

- Ask the tutor to explain your own schematic, sheet by sheet
- Ask it to find your ERC errors — then read them yourself
- Autoroute one channel, then look hard at what it did
- On your own gapped project, your own zone, your own branch
- Never on main, and never on someone else's section
- The class rule: your copper is yours and you can explain it



place them back, then ERC

- Open your gapped sheets — each is its own small KiCad project
- Place the missing parts back: symbol, value, footprint field
- Keep the reference designators the PDF shows
- Run ERC — the dangling ends where parts were missing must go
- Then route inside your zone, and run DRC often
- Decoupling next to the pin; no tracks on the inner layers

- Branch, commit and push from VS Code's Source Control panel
- Open a pull request with your 3D screenshot in the description
- Request your ring reviewer: A reviews B, B reviews C, C reviews A
- Leave at least two comments, each against a checklist line
- Approve, or request changes — then fix your own
- The automatic build must be green before it can merge

SCREENSHOT — A GitHub pull request on class-board-2026 from a student branch: the conversation view with a review comment quoting a line of the review checklist, the requested reviewer visible, the checks section green, and the Approve / Request changes controls.

- From Workshop 3: one driver and one phone panel per section
- Your Project 2 app grows into the instrument's front end
- Start from the class reference firmware — a known good baseline
- That is what separates a hardware problem from a software one
- Which of the five projects is yours?
- Can you adapt it to solve a problem in your research?

SCREENSHOT — The class reference web app in a phone-width browser window, showing its tab bar and one tab open (for example the Scope or Status tab) with live values.

- Before the next workshop: complete the preparation, improve your apps, build and have fun.
- The preparation is what the factory order needs from you
- Finish the routing: DRC 0 errors, 0 unrouted, pour joined
- Fetch your one part from the JLCPCB parts library
- Open your pull request; review your neighbour's
- Gerbers go to the factory Mon 28 Sep, 2 pm — hard cutoff

SCREENSHOT — The project wall (showcase-2026) on the projector, full screen, self-refreshing view with the class's project cards.

File	Title	License	Author
pick_and_place_1.jpg	Pick and place internals of surface mount machine	CC BY-SA 3.0	Peripitus
reflow_oven_1.jpg	Reflow oven	CC BY-SA 3.0	Nelatan
stencil_paste_b_1.jpg	Solder printing machine	Public domain	Nelatan
wave_soldering_1.jpg	Seho MWS 2340	CC BY-SA 2.0 de	Pedro ximenez
smd_chip_closeup_1.jpg	Two JRC RC4580 dual op amps on a sound card PCB	CC BY-SA 3.0	Samtoucan
smd_components_1.jpg	Sony-CCD-TRV24E sensor 343f	CC BY 4.0	Thomas Bresson
chip_tray_reel_1.jpg	SMD capacitors in tape	CC BY-SA 4.0	Filip Wahlberg
chip_die_1.jpg	07R01	CC BY-SA 4.0	Mister rf
chip_die_3.png	TI AHC08 Die Shot (4x2-Input AND Gate)	CC BY-SA 2.0	Cole L
logic_gates_1.jpg	7400 Series Chips 1	CC BY-SA 3.0	JailsonBR
fpga_board_1.jpg	Sipeed Tang Nano 9K FPGA	CC0	Popolon
bare_pcb_b_1.jpg	Macro photograph of the RP2350 microcontroller on a Raspberry Pi Pico 2 board	CC0	Profpcde
raspberry_pi_1.jpg	Raspberry Pi 4 Model B - Side	CC BY-SA 4.0	Laserlicht
esp32_module_1.jpg	ESP32-C3 with Traffic Light Module from Above	CC0	DavesCodeMusings
sma_panel_1.jpg	Male 50 ohm SMA connector	CC BY-SA 3.0	Swift.Hg
oscilloscope_1.jpg	Screen of Hameg Oscilloscope HMO3004 Series	CC0	Ordercrazy
kicad_screenshot_1.png	KiCad-Pcbnew-Hackrf-One	CC BY-SA 3.0 de	Philntech
soldering_iron_2.jpg	Hobbies (24443540461)	CC BY-SA 2.0	cchana

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